

CareBridge

Complete User & Technical Documentation
AI Knowledge Base · RAG Training Corpus · FAQ Repository

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A need-driven donation and visit-scheduling platform connecting verified donors with verified orphanages.

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Chapter 1 — Introduction

1.1 About CareBridge

CareBridge is a need-driven donation and visit-scheduling platform that connects verified donors with verified orphanages through a transparent and structured ecosystem. Rather than asking donors to give generically, CareBridge lets orphanages publish exactly what they need — a category of item, a quantity, a price per unit, and a priority level — so that every donation is targeted at a real, stated requirement. The platform also formalizes the process of visiting an orphanage in person, replacing ad-hoc phone calls and informal arrangements with a structured slot-booking and confirmation workflow.

The system is composed of three cooperating applications: a Spring Boot backend API that owns all business logic and data, a React-based donor/orphanage frontend that the public uses day to day, and a separate React-based admin portal used exclusively by platform administrators to verify new donors and orphanages. This separation keeps the trust-and-safety workflow isolated from the consumer-facing experience while both sides talk to the same backend API.

1.2 Problem Statement

Individuals willing to support orphanages typically face several structural obstacles. There is limited visibility into what an orphanage actually needs at any given moment, which leads to donations that are well-intentioned but mismatched to real requirements. It is difficult for an ordinary donor to independently verify that an organization calling itself an orphanage is legitimately registered and operating. Once a donation is made, there is often no visibility into whether or how it was used. The overall donation process is unstructured — conducted over phone calls, word of mouth, or social media — and there is no organized system for scheduling an in-person visit, which can result in overcrowding, missed appointments, or orphanages being unable to plan around visitor volume.

CareBridge addresses each of these problems directly: verification (via PAN, DARPAN, and JJ Act Certificate review by an admin) establishes legitimacy before an orphanage can participate; need items make requirements explicit and quantified; donation status tracking (PENDING → DELIVERED) and delivery photo proof create an audit trail; and the slot/booking system gives orphanages control over how many visitors arrive and when.

1.3 Mission

- Connect verified donors and orphanages within a single trusted ecosystem.
- Enable orphanages to publish their actual, current requirements rather than relying on generic appeals.
- Facilitate targeted, transparent donations that map directly to a stated need.
- Enable structured visit scheduling that respects both donor intent and orphanage capacity.
- Track donation fulfillment and impact so that progress toward a need is always visible.
- Improve accountability and efficiency in charitable giving broadly.

1.4 Vision

CareBridge envisions a giving ecosystem in which trust is established structurally — through verified identity and documentation — rather than through personal relationships or reputation alone, and in which every donor can see, in concrete terms, the difference their contribution made. As the roadmap in Chapter 9 outlines, this vision extends toward richer analytics, corporate giving programs, and eventually blockchain-backed transparency for donation records.

1.5 Objectives

- Provide a single source of truth for what each orphanage currently needs, organized by category and priority.
- Reduce the friction of scheduling an in-person donation visit through a formal slot system.
- Give donors a dashboard view of their cumulative impact rather than a one-off receipt.
- Give orphanages a dashboard view of fulfillment progress across all posted needs.
- Maintain an administrative gatekeeping step so that only verified organizations can receive donations.
- Keep the system auditable by archiving rejected profiles and deleted items rather than discarding them.

1.6 Why CareBridge Was Built

CareBridge was conceived from direct exposure to the practical difficulties of supporting orphanages informally: not knowing what was actually needed, not being certain a given organization was legitimate, and having no structured way to arrange a visit. The platform formalizes each of those informal, trust-dependent steps into an explicit workflow backed by verification, tracked state, and notifications.

1.7 Platform Overview

At a glance, CareBridge consists of three portals sharing one backend: the **Donor Portal**, where individuals register, get verified, browse needs, donate, and book visits; the **Orphanage Portal**, where organizations register, get verified, post needs, receive donations, and manage visit slots; and the **Admin Portal**, where platform staff review and approve or reject incoming donor and orphanage profiles. A public-facing **Landing Page** layer exposes aggregate statistics and browsable content to visitors who have not yet registered, and an AI chatbot (**CareBot**) is embedded in the donor/orphanage frontend to answer common questions (see Chapter 5).

1.8 Core Values

Accountability — Enforced through mandatory admin verification and document requirements for every donor and orphanage.

Transparency — Need posting, donation status tracking, and impact metrics make the state of every transaction visible to both parties.

Efficiency — Need items with categories and priorities let donors match their giving to what is actually required.

Trust — Profile verification and a permanent rejection/archival trail underpin confidence in every counterparty.

User Experience — Purpose-built interfaces for donors, orphanages, and admins, each tailored to that role's workflow.

1.9 Benefits

For Donors

- See exactly what an orphanage needs before giving.
- Choose between in-person delivery and online-ordered delivery.
- Track cumulative impact across all past donations on one dashboard.
- Schedule visits without relying on informal phone calls.

For Orphanages

- Publish specific, quantified needs instead of generic appeals.
- Control visitor volume through slot capacity limits.
- See fulfillment progress on every posted need in real time.
- Build credibility through a documented, admin-verified profile.

1.10 Real-World Use Cases

A first-time donor browsing before committing. A visitor lands on the public homepage, checks GET /api/landing/stats and GET /api/landing/orphanages to see verified organizations and impact numbers, then decides to register once satisfied the platform is legitimate.

An orphanage running low on school uniforms. The orphanage posts a NeedItem in the EDUCATION category with HIGH priority and a quantity of 100. As donors contribute, fulfilledQuantity and reservedQuantity climb toward that total, visible on both the orphanage dashboard and the public needs feed.

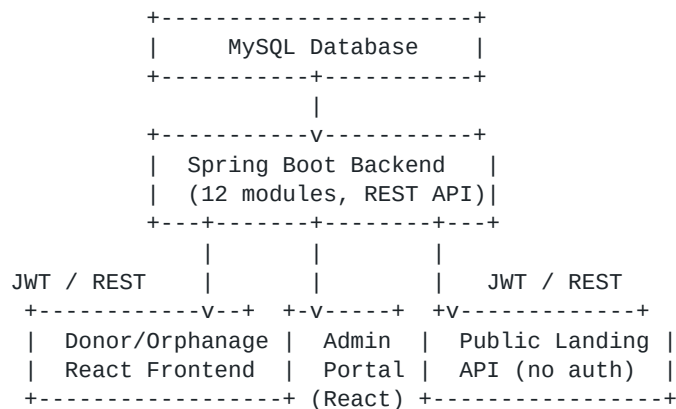
A donor combining an online order with a future visit. A donor places an ONLINE_ORDER donation for medical supplies with a tracking number, and separately books a visit slot for the following month to deliver additional items in person — the two donation types coexist independently.

An admin protecting the platform from a fraudulent orphanage signup. An admin reviewing a pending orphanage profile notices the PAN photo does not match the DARPAN registration details and rejects the profile with a specific reason, which is emailed to the applicant and archived in RejectedOrpProfile.

Chapter 2 — Understanding the Platform

2.1 Platform Architecture

CareBridge follows a layered, modular monolith design on the backend (Controller → Service → Repository → Entity) built with Spring Boot 4.0.6 on Java 21, backed by MySQL in production and H2 for testing. Twelve functional modules divide responsibility cleanly: authentication, donor profile management, orphanage profile management, donation management, visit-booking management, needs management, dashboard management (donor and orphanage variants), notification management, admin management, slot-details fetch management, landing-page data management, and shared common utilities. Two independent React single-page applications — the main frontend and the admin portal — consume this API over REST, authenticating with JWT bearer tokens.



2.2 User Roles

Donor. A registered, PAN-verified individual who browses needs, donates items (in-person or online-ordered), and books orphanage visits.

Orphanage. A registered, document-verified organization (PAN, DARPAN, JJ Act Certificate) that posts needs, receives donations, and manages visit slots.

Administrator. Platform staff who review pending donor and orphanage profiles and approve or reject them, gatekeeping who may transact on the platform.

CareBot. An AI chatbot embedded in the frontend that answers common questions about registration, donations, verification, and bookings using a retrieval-augmented knowledge base (see Chapter 5 and Chapter 8).

2.3 Interaction Flow

The typical interaction between a donor and an orphanage flows through several shared system components rather than direct contact: the donor discovers a need item posted by the orphanage, submits a donation request referencing that need, and the orphanage confirms delivery once the item arrives. For in-person giving, the orphanage's published slots and the donor's booking request pass through a

confirm/reject decision before a visit is scheduled. Every step in this flow triggers an email notification to keep both parties informed without requiring direct communication.

2.4 Platform Lifecycle

1. Registration & email verification (OTP) for both donors and orphanages.
2. Detailed profile creation with identity/registration documents.
3. Admin review and verification (approve to VERIFIED or reject with reason).
4. Orphanages post need items; donors browse and donate against them.
5. Optional visit-slot booking and confirmation for in-person donations.
6. Delivery confirmation by the orphanage, closing the donation lifecycle.
7. Ongoing dashboard reporting of impact and fulfillment on both sides.

Chapter 3 — Complete Feature Documentation

This chapter documents every major feature of CareBridge in a consistent format: Overview, Purpose, Workflow, Benefits, User Experience considerations, Best Practices, Common Errors, and representative FAQs. Each section is grounded strictly in the CareBridge knowledge base; where the knowledge base does not specify a detail, that gap is called out explicitly rather than filled in with an assumption (see also Chapter 7, Knowledge Gap Analysis).

3.1 Registration

Overview. The entry point for creating a CareBridge account, gated by email OTP verification before any user record is persisted.

Purpose. Ensure every account on the platform is tied to a real, reachable email address before granting access to any further feature.

Workflow

- POST /auth/register accepts name, email, and password, and emails a six-digit OTP.
- POST /auth/verify accepts that OTP and, on success, creates the User record with role DONOR, ORPHANAGE, or ADMIN.
- POST /auth/resend issues a fresh OTP if the original is lost or expires.

Benefits

- Filters out invalid or mistyped email addresses before they enter the system.
- Keeps the registration data model minimal — full profile detail is captured separately.

User Experience Notes

- The registration form should separate 'create account' from 'complete profile' visually, so users understand these are two distinct steps.

Best Practices

- Never create the User record before OTP verification succeeds, to avoid orphaned unverified accounts.

Common Errors

- UserAlreadyRegisteredException when the email is already taken.
- InvalidOTPException or OtpExpiredException on a bad or stale code.

Frequently Asked Questions

Q: Do I need to verify my email before creating my donor/orphanage profile?

A: Yes — registration and OTP verification must complete first; profile creation is a separate subsequent step.

3.2 Email Verification & OTP

Overview. The One-Time Password mechanism underlying registration, password reset, and orphanage verification across the platform.

Purpose. Provide a lightweight, time-boxed proof of email ownership without requiring a persistent third-party identity provider.

Workflow

- A random six-digit OTP is generated and stored with a five-minute expiry and a used flag.
- The OTP is emailed via the configured SMTP provider (Gmail in dev, Brevo in production).
- On submission, the OTP is checked for correctness, expiry, and prior use before being marked used.

Benefits

- Five-minute expiry limits the window in which an intercepted OTP could be exploited.
- Single-use enforcement prevents replay of an already-consumed code.

User Experience Notes

- A visible countdown timer on the OTP screen reduces the chance of users waiting past expiry before submitting.

Best Practices

- Cap the number of resend attempts to prevent OTP-spamming a single inbox or abusing the email pipeline.

Common Errors

- `OtpNotFoundException` if no OTP record exists for the submitted email.
- `OtpAlreadyUsedException` if a previously consumed code is resubmitted.

Frequently Asked Questions

Q: How long is an OTP valid?

A: Five minutes from generation, after which it must be resent.

3.3 Forgot Password

Overview. A self-service password recovery flow built on the same OTP mechanism used elsewhere in the platform.

Purpose. Let users regain account access without administrator intervention while still proving ownership of the registered email.

Workflow

- POST /auth/forgot-password emails an OTP to the account's registered address.
- POST /auth/verify-forgot-otp checks that OTP's validity.
- POST /auth/reset-password accepts and stores a new password once the OTP step succeeds.

Benefits

- Removes password-reset support burden from platform administrators.
- Reuses the same secure OTP infrastructure rather than a separate, parallel mechanism.

User Experience Notes

- Reset-password forms should enforce the same password-strength rules shown during initial registration, for consistency.

Best Practices

- Invalidate any existing session/token context tied to the account once a password reset succeeds, if session invalidation is supported.

Common Errors

- Submitting an expired forgot-password OTP requires restarting the flow from POST /auth/forgot-password.

Frequently Asked Questions

Q: Will I be logged out of other devices after resetting my password?

A: The knowledge base does not document session-invalidation behavior on password reset — flagged as a gap in Chapter 7.

3.4 Login & Session Handling

Overview. Authenticates a user and issues the JWT bearer token used to authorize every subsequent protected request.

Purpose. Provide access without server-side session storage, keeping the backend stateless and horizontally scalable.

Workflow

- POST /auth/login validates email and password, returning a JWT valid for 24 hours on success.
- The frontend stores this token (via authUtils.js) and attaches it as an Authorization: Bearer header on every protected request.
- JwtAuthFilter validates the token's signature and expiry on the backend for every incoming request.

Benefits

- No server-side session store to scale, replicate, or invalidate across instances.
- A single, well-understood token format simplifies both frontend and backend logic.

User Experience Notes

- An expired-session redirect to login should preserve the user's intended destination so they land back where they left off after re-authenticating.

Best Practices

- Never store the JWT in a way that's readable by injected third-party scripts if that becomes a concern; document the storage mechanism clearly for the security team.

Common Errors

- HTTP 401 on any protected endpoint once the 24-hour token window has elapsed.

Frequently Asked Questions

Q: Can I stay logged in longer than 24 hours?

A: Not with the current token expiry; you would need to log in again once it lapses.

3.5 Donor Profile Management

Overview. Captures the personal, contact, and identity-verification details required before a donor may transact on the platform.

Purpose. Give admins the information needed to verify a donor is a real, identifiable individual, and give the platform a stable CareBridgeID to reference in every future transaction.

Workflow

- POST /donor/create-profile submits name, DOB, designation, phone, PAN number, and PAN photo (plus optional address and profile picture).
- The profile is created with status PENDING and a generated CareBridgeID.
- An admin approves (VERIFIED) or rejects (with reason, archived) the profile.
- PUT /donor/update-profile allows edits before or after verification.

Benefits

- A single durable CareBridgeID ties together donations, bookings, and dashboard data.
- PAN-based verification adds real-world accountability to an otherwise anonymous internet interaction.

User Experience Notes

- Multipart file upload for PAN photo should show a clear preview before submission.

- Validation errors (e.g., malformed PAN) should surface inline, not just after submission.

Best Practices

- Validate PAN format (5 letters, 4 digits, 1 letter) client-side before upload to save round-trips.
- Keep optional fields genuinely optional in the UI to reduce signup abandonment.

Common Errors

- Duplicate phone number across profiles triggers a uniqueness violation.
- Missing PAN photo blocks profile submission entirely, since it is a required field.

Frequently Asked Questions

Q: Can I submit my profile without a PAN photo?

A: No, PAN photo is a required upload for donor profile creation.

Q: Does editing my profile reset my VERIFIED status?

A: No, simple field updates do not require re-approval.

3.6 Orphanage Profile Management

Overview. Captures organizational identity and statutory compliance documents for orphanages seeking to post needs and receive donations.

Purpose. Ensure only legitimate, registered organizations can appear on the platform and accept donations from the public.

Workflow

- POST /orphanage/create-profile submits orphanage name, admin name, number of children, contact details, DARPAN ID, PAN number/photo, and JJ Act Certificate photo.
- POST /orphanage/verifyOrpOTP confirms the registration email via OTP.
- An admin reviews documents and approves (VERIFIED) or rejects (archived, with reason emailed) the profile.
- PUT /orphanage/update-profile allows subsequent edits.

Benefits

- Statutory-document verification (DARPAN, JJ Act) is a stronger trust signal than self-reported information alone.
- A verified orphanage profile becomes discoverable on the public landing page, extending its reach to potential donors.

User Experience Notes

- Because four separate documents are required, a multi-step wizard reduces abandonment compared to one long form.

- Clear labeling of DARPAN and JJ Act Certificate (unfamiliar terms to many users) helps first-time registrants.

Best Practices

- Cross-check DARPAN ID format before accepting a submission to catch obvious typos early.
- Keep rejection reasons specific and actionable, since resubmission requires a full new registration.

Common Errors

- Missing JJ Act Certificate photo blocks submission, since it is required.
- Duplicate orphanage phone number triggers a uniqueness violation.

Frequently Asked Questions

Q: Is DARPAN registration mandatory?

A: Yes, it is a required field for orphanage profile creation.

Q: Can an orphanage operate before verification?

A: No, only VERIFIED orphanages can post needs and receive donations.

3.7 Need Item Management

Overview. Lets orphanages publish specific, quantified requirements that donors can browse and contribute toward.

Purpose. Replace generic donation appeals with concrete, trackable requirements organized by category and priority.

Workflow

- POST /orphanage/needItem/create-item defines name, category, quantity, price per unit, priority, and optional description/product links.
- Donors browse via GET /donor/allNeeds, filterable by category and priority.
- As donations are made, fulfilledQuantity and reservedQuantity update; remaining quantity is total minus fulfilled minus reserved.
- PUT /orphanage/needItem/updateItem/{id} and DELETE /orphanage/needItem/deleteItem/{id} manage the item's lifecycle.

Benefits

- Category and priority fields let donors quickly find needs matching their interest or urgency preference.
- Fulfillment tracking prevents a need from being over-committed beyond its actual quantity.

User Experience Notes

- Progress bars showing fulfilled vs. remaining quantity communicate status at a glance.

- Product links help donors purchasing online find a matching item quickly.

Best Practices

- Keep need descriptions specific (e.g., sizes for uniforms) to reduce mismatched donations.
- Archive rather than hard-delete needs so historical fulfillment data is preserved (see DeletedItems).

Common Errors

- Attempting to donate more than the remaining quantity should be rejected server-side.
- Editing needItemId or the owning orphanage ID is not permitted.

Frequently Asked Questions

Q: What happens when a need is fully funded?

A: It can remain active but shows zero remaining quantity until archived.

Q: Can donors see who else has committed to a need?

A: The knowledge base does not document donor-to-donor visibility on a shared need — flagged in Chapter 7.

3.8 Donation Management

Overview. Handles the full lifecycle of a donation, from creation through delivery confirmation, across both in-person and online-order donation types.

Purpose. Provide a trackable, auditable record connecting a donor's contribution to a specific need item and its eventual delivery.

Workflow

- POST /donor/donate creates a donation (PENDING) referencing a needItemId, quantity, and donation type with type-specific fields.
- Donors can view, update (PATCH /donor/update/{id}), or delete (DELETE /donor/delete-pending-donation/{id}) donations while PENDING.
- The receiving orphanage confirms delivery via POST /orphanage/delivered/{id}, moving status to DELIVERED and recording a note and photo proof.
- Email notifications fire at creation, delivery, and cancellation.

Benefits

- Two donation types (IN_PERSON, ONLINE_ORDER) accommodate different donor circumstances without forcing a single channel.
- Delivery photo proof builds trust for both the donor and future donors viewing that orphanage's history.

User Experience Notes

- Donation forms should adapt their required fields dynamically based on the selected donation type.
- Status badges (PENDING/DELIVERED/CANCELLED) should be color-coded for quick scanning.

Best Practices

- Validate requested quantity against remaining need quantity before accepting a donation to avoid over-commitment.
- Require a delivery photo for online-order confirmations to preserve an audit trail comparable to in-person visits.

Common Errors

- Trying to update or cancel a DELIVERED donation is not permitted.
- Missing type-specific fields (e.g., trackingId for ONLINE_ORDER) blocks creation.

Frequently Asked Questions

Q: Can I split one donation across two need items?

A: No — each donation request references exactly one needItemId; make a separate donation for a different need.

Q: What proof does an online-order donation require?

A: A platform name, tracking ID, and an order proof image.

3.9 Visit Booking & Slot Management

Overview. Lets orphanages publish available visit time windows and lets donors reserve a spot within one, subject to capacity and verification checks.

Purpose. Replace informal visit arrangements with a structured, capacity-aware booking system that both parties can rely on.

Workflow

- POST /orphanage/slot/create defines date, time window, and visitor capacity.
- GET /donor/slots/available-slots/{orphanageId} lists AVAILABLE slots for browsing.
- POST /donor/slots/book reserves a slot for a validated, VERIFIED donor.
- The orphanage confirms (PATCH /orphanage/slot/confirm/{id}) or rejects (PATCH /orphanage/slot/reject/{id}) each booking.
- SlotSchedulerService automatically marks past slots EXPIRED.

Benefits

- Capacity limits prevent orphanages from being overwhelmed by visitors.
- Automatic expiry keeps the available-slots list clean without manual maintenance.

User Experience Notes

- A calendar-style view of available slots is more scannable than a flat list.
- Booking confirmation state should be visible immediately, not only via email.

Best Practices

- Run the slot-expiry scheduler frequently enough that stale slots don't linger as 'available' after their time has passed.
- Surface rejection reasons to donors so they understand why a booking was declined.

Common Errors

- Booking more visitors than remaining capacity is rejected.
- Booking as an unverified donor is rejected.

Frequently Asked Questions

Q: Do slots repeat weekly automatically?

A: No — each slot is a one-off window; orphanages must create new slots for recurring availability.

Q: Can I book two different slots for the same day?

A: The knowledge base does not explicitly restrict this — flagged in Chapter 7.

3.10 Dashboards & Analytics

Overview. Provides role-specific summaries of activity, fulfillment, and impact for donors and orphanages.

Purpose. Turn raw transactional data into an at-a-glance view of progress and impact for both sides of the platform.

Workflow

- Donor dashboard aggregates profile status, donation statistics, impact summary, recent donations, upcoming bookings, and recommended urgent needs.
- Orphanage dashboard aggregates an overview, donation summary, need summary, visit summary, and activity feed.

Benefits

- Impact metrics (e.g., estimated lives impacted) give donors a tangible sense of their contribution's effect.
- Orphanage-side fulfillment percentages help prioritize which needs to promote next.

User Experience Notes

- Dashboards should load progressively (summary first, detail on demand) rather than blocking on every metric at once.
- Recent-activity feeds benefit from relative timestamps ('2 days ago') over raw dates.

Best Practices

- Cache expensive aggregate queries where the underlying data changes infrequently.
- Keep impact-estimate methodology consistent so numbers remain comparable over time.

Common Errors

- Displaying stale cached statistics after a fresh donation can confuse users if cache invalidation is missed.

Frequently Asked Questions

Q: How often does dashboard data update?

A: The knowledge base does not specify a refresh interval — flagged as a gap in Chapter 7.

Q: Can orphanages see individual donor dashboards?

A: No — dashboards are private to the profile that owns them.

3.11 Notifications (Email & OTP)

Overview. Delivers OTPs and transactional notifications by email at every key state transition across the platform.

Purpose. Keep both donors and orphanages informed without requiring them to poll the platform manually.

Workflow

- OTP generation, storage (5-minute expiry, single-use), and validation underpin registration, password reset, and orphanage verification.
- EmailMessages.java supplies formatted templates for OTPs, approvals, rejections, donation events, and booking decisions.
- Gmail SMTP is used in development; Brevo SMTP API is used in production.

Benefits

- A consistent template library keeps notification tone and formatting uniform across the platform.
- Single-use, time-boxed OTPs reduce the risk of stale codes being reused.

User Experience Notes

- Emails should clearly state the next action required (e.g., 'Enter this code within 5 minutes').
- Rejection emails should state the reason plainly, since users cannot resubmit the same profile.

Best Practices

- Rate-limit OTP resend requests to prevent abuse of the email pipeline.
- Keep SMTP credentials strictly in environment variables, never in source control.

Common Errors

- OTP emails landing in spam is a frequent real-world support issue (see Chapter 6).

Frequently Asked Questions

Q: Can I choose SMS instead of email for OTPs?

A: Not currently — email is the only OTP channel documented; SMS is a future-roadmap item.

3.12 Messaging

Overview. An in-platform messaging surface allowing donors and orphanages to communicate directly.

Purpose. Give both parties a channel for donation- or visit-related questions without exchanging personal contact information.

Workflow

- DonorMessages.jsx and OrphanageMessagePage.jsx provide role-specific message views backed by a shared Messages.jsx component.

Benefits

- Keeps donor/orphanage communication on-platform, where context (donations, bookings) is already available.

User Experience Notes

- Message threads should be linkable to the relevant donation or booking for context.

Best Practices

- Moderate or flag messaging content if abuse becomes a concern, particularly given the platform serves vulnerable populations.

Common Errors

- The knowledge base does not document real-time delivery (e.g., WebSockets); behavior beyond basic message display is a documentation gap (Chapter 7).

Frequently Asked Questions

Q: Is messaging real-time?

A: The knowledge base does not confirm real-time delivery; real-time chat via WebSockets is listed as a near-term roadmap item, suggesting the current implementation may not be real-time.

3.13 Security (Authentication, RBAC, CORS)

Overview. The set of protections governing who can access which endpoint and how credentials and tokens are handled.

Purpose. Protect user data and platform integrity while keeping the authentication model simple and stateless.

Workflow

- JwtAuthFilter validates every request's bearer token and populates the SecurityContext.
- SecurityConfig defines public routes (/auth/**, /api/landing/**, Swagger) and role-restricted routes (ROLE_DONOR, ROLE_ORPHANAGE, ROLE_ADMIN).
- CORS configuration whitelists specific frontend origins.

Benefits

- Stateless JWT auth scales horizontally without shared session storage.
- Clear role segmentation prevents a donor account from reaching admin or orphanage-only endpoints.

User Experience Notes

- Expired-session handling should redirect to login gracefully rather than showing a raw error.

Best Practices

- Rotate the JWT signing secret periodically and store it only in environment variables.
- Apply the principle of least privilege when adding new endpoints to role rules.

Common Errors

- A misconfigured CORS origin list is the most common integration issue between frontend and backend (see Chapter 6).

Frequently Asked Questions

Q: Is two-factor authentication available?

A: Not yet — it is listed as a security practice still to be implemented (Chapter 10).

3.14 Landing Page & Public Browsing

Overview. Unauthenticated endpoints that expose aggregate statistics and browsable content to visitors who have not yet registered.

Purpose. Let prospective donors and orphanages evaluate the platform's legitimacy and impact before creating an account.

Workflow

- GET /api/landing/stats returns global metrics (verified orphanages, active donors, donations received, items distributed, success stories).
- GET /api/landing/orphanages and GET /api/landing/needs expose browsable, verified content publicly.

- GET /api/landing/features returns platform feature descriptions.

Benefits

- Public statistics build credibility with first-time visitors before any signup friction is introduced.

User Experience Notes

- Public pages should load fast and cache aggressively since the underlying data changes relatively slowly.

Best Practices

- Ensure only VERIFIED profiles are aggregated into public statistics to avoid exposing unverified or rejected data.

Common Errors

- None specific to this module are documented in the knowledge base.

Frequently Asked Questions

Q: Do I need an account to see impact numbers?

A: No — the landing statistics endpoints are public and require no authentication.

3.15 Admin Review & Verification

Overview. The administrative workflow for approving or rejecting donor and orphanage profiles, gatekeeping platform participation.

Purpose. Ensure every participant has been manually vetted before they can transact.

Workflow

- GET /admin/donor/pending and GET /admin/orphanage/pending list queues for review.
- PATCH ../approve/{id} sets status to VERIFIED and emails confirmation.
- PATCH ../reject/{id} accepts a reason, archives the profile (RejectedDonorProfile / RejectedOrpProfile), deletes the original record, and emails the reason.

Benefits

- A manual human review step catches nuanced red flags that automated checks might miss.
- Archival of rejected profiles preserves an audit trail without cluttering the active dataset.

User Experience Notes

- Document viewers in the admin portal should support zoom, since PAN and certificate photos need close inspection.

Best Practices

- Keep rejection reasons specific and consistent to reduce repeat invalid resubmissions.
- Track average review time to identify bottlenecks in the approval queue.

Common Errors

- A profile stuck in PENDING for a long time typically indicates the admin queue has not reached it yet, not a system failure.

Frequently Asked Questions

Q: Can one admin account handle both donor and orphanage reviews?

A: Yes — AdminDonorController and AdminOrpController are separate controllers but reviewed by the same admin role.

Chapter 4 — Complete User Workflows

This chapter walks through the complete, step-by-step journeys for each of CareBridge's three user roles, from first contact with the platform through ongoing usage.

4.1 Donor Workflow

Step 1 — Register. Submit name, email, and password to POST /auth/register; receive an OTP by email.

Step 2 — Verify Email. Submit the OTP to POST /auth/verify to complete account creation.

Step 3 — Log In. Authenticate via POST /auth/login to receive a 24-hour JWT.

Step 4 — Create Profile. Submit personal details, PAN, and PAN photo to POST /donor/create-profile; status becomes PENDING.

Step 5 — Await Admin Approval. An admin reviews the profile and sets status to VERIFIED (or REJECTED with a reason).

Step 6 — Browse Needs. Use GET /donor/allNeeds to explore active needs across verified orphanages, filterable by category and priority.

Step 7 — Donate. Submit POST /donor/donate referencing a needItemId, choosing IN_PERSON or ONLINE_ORDER.

Step 8 — Schedule a Visit (optional). Browse available slots and book one via POST /donor/slots/book if donating in person.

Step 9 — Track Donations. Monitor pending, delivered, and cancelled donations via the dashboard and dedicated list endpoints.

Step 10 — Review Impact. Check the donor dashboard for cumulative statistics and estimated impact.

4.2 Orphanage Workflow

Step 1 — Register. Submit organizational and contact details to POST /orphanage/create-profile, including DARPAN ID, PAN, and JJ Act Certificate.

Step 2 — Verify Email. Confirm the OTP sent to the orphanage's email via POST /orphanage/verifyOrpOTP.

Step 3 — Await Admin Approval. An admin reviews the submitted documents and sets status to VERIFIED (or REJECTED with a reason).

Step 4 — Post Needs. Use POST /orphanage/needItem/create-item to publish specific, quantified requirements by category and priority.

Step 5 — Create Visit Slots. Publish available visiting windows via POST /orphanage/slot/create, including date, time, and capacity.

Step 6 — Receive Donations. Monitor incoming PENDING donations via GET /orphanage/pending-donations.

Step 7 — Manage Bookings. Review pending visit bookings and confirm or reject each one.

Step 8 — Confirm Deliveries. Once an item arrives, call POST /orphanage/delivered/{id} with a note and photo proof.

Step 9 — Monitor Dashboard. Track fulfillment percentage, donation summary, and visit summary on the orphanage dashboard.

4.3 Administrator Workflow

Step 1 — Log In. Authenticate through the dedicated admin login page.

Step 2 — Review Pending Donor Profiles. Inspect submitted PAN details and photos via GET /admin/donor/pending.

Step 3 — Approve or Reject Donors. Call the approve or reject endpoint, supplying a reason if rejecting.

Step 4 — Review Pending Orphanage Profiles. Inspect DARPAN, PAN, and JJ Act Certificate documents via GET /admin/orphanage/pending.

Step 5 — Approve or Reject Orphanages. Call the approve or reject endpoint, supplying a reason if rejecting.

Step 6 — Monitor Donations & Visits. Use the admin dashboard to track platform-wide donation and booking activity.

Step 7 — Maintain System Health. Review statistics such as pending-approval counts and recent activity.

Chapter 5 — Chatbot Knowledge Base (CareBot)

This chapter forms the primary retrieval corpus for CareBot, CareBridge's AI chatbot. Each entry below documents one real user intent grounded in the CareBridge knowledge base, expressed as a canonical question plus several natural, casual, typo, and voice-search style rephrasings that a real user might type or speak. Every rephrasing maps to the same answer, intent, category, and metadata, which is the correct chunking strategy for a RAG system: variants should route to one canonical answer rather than each being treated as a new fact (see Chapter 8 for embedding and chunking guidance).

Entries are grouped by category. Each entry lists: the canonical question, detected intent, the grounded answer, alternative phrasings (including a voice-search style and a typo variant), likely follow-up questions, related topic IDs, keywords for lexical/BM25-style retrieval, the primary user persona, and a difficulty rating.

5.x Registration & Onboarding

[REG-01] Q: How do I register on CareBridge?

Intent: Informational — registration & onboarding

Answer: Registration starts at POST /auth/register, which generates a One-Time Password (OTP) and emails it to the address you supplied. You then call POST /auth/verify with that OTP to confirm your identity and create your user account. If the OTP does not arrive in time, POST /auth/resend issues a new one (a five-minute expiry window applies to each OTP).

Alternative phrasings: quick q - how do I register on CareBridge?; How do I registr on CareBridge?; hey carebot, how do i register on carebridge; register?

Possible follow-ups: What happens if I do not receive the OTP email? | Do I need to create a profile after registering?

Related topics: REG-02, AUTH-01, PROFILE-01 **Keywords:** register, sign up, create account, OTP **Persona:** New Visitor **Difficulty:** Basic

[REG-02] Q: Is registering an account the same as creating my donor or orphanage profile?

Intent: Informational — registration & onboarding

Answer: No. Registration (POST /auth/register + POST /auth/verify) only creates a basic User record with a name, email, hashed password, and role (DONOR, ORPHANAGE, or ADMIN). A separate step — POST /donor/create-profile or POST /orphanage/create-profile — captures the detailed information (PAN, address, documents, etc.) needed before you can use the platform.

Alternative phrasings: hey, is registering an account the same as creating my donor or orphanage profile?; Is registering an account teh same as creating my donor or orphanage profile?; hey carebot, is registering an account the same as creating my donor or orphanage profile; account vs profile?

Possible follow-ups: What information does the profile step ask for? | Can I use the platform right after registering?

Related topics: REG-01, PROFILE-01, PROFILE-05 **Keywords:** account vs profile, registration steps, difference **Persona:** New Visitor **Difficulty:** Basic

[REG-03] Q: Why did my registration fail?

Intent: Informational — registration & onboarding

Answer: The most common causes are: the email is already registered (emails must be unique per account), the email format is invalid, the password does not meet strength requirements, a temporary server/connection issue occurred, or a form validation rule was not met. Checking the on-screen validation messages usually identifies the exact field that needs correcting.

Alternative phrasings: hey, why did my registration fail?; Why did my registration fail?; hey carebot, why did my registration fail; registration failed?

Possible follow-ups: What if the email is already taken? | Where can I see which field failed validation?

Related topics: REG-01, TRBL-01 **Keywords:** registration failed, cant register, sign up error **Persona:** New Visitor
Difficulty: Basic

[REG-04] Q: How long is the OTP valid for during registration?

Intent: Informational — registration & onboarding

Answer: Every OTP issued by CareBridge, including the one sent during registration, expires five minutes after it is generated. If it lapses, request a new code through the resend endpoint rather than retrying the expired one.

Alternative phrasings: just wondering, how long is the OTP valid for during registration?; How long is the OTP valid for during registration?; hey carebot, how long is the otp valid for during registration; otp expiry?

Possible follow-ups: How many times can I resend an OTP? | What happens if I enter an expired OTP?

Related topics: OTP-01, OTP-02 **Keywords:** otp expiry, otp validity, five minutes **Persona:** New Visitor **Difficulty:** Basic

[REG-05] Q: Does orphanage registration require OTP verification too?

Intent: Informational — registration & onboarding

Answer: Yes. After submitting POST /orphanage/create-profile, the orphanage must call POST /orphanage/verifyOrpOTP to confirm the OTP emailed to the orphanage's registered address. POST /orphanage/resendOrpOTP is available if the original OTP is not received or expires.

Alternative phrasings: hey, does orphanage registration require OTP verification too?; Does orphanage registration require OTP verification too?; hey carebot, does orphanage registration require otp verification too; orphanage otp?

Possible follow-ups: What if the orphanage OTP expires before I verify it? | What happens after OTP verification — is the orphanage automatically active?

Related topics: REG-01, PROFILE-05, VERIF-02 **Keywords:** orphanage otp, orphanage verification, orp signup **Persona:** Orphanage Admin **Difficulty:** Intermediate

5.x Authentication & Security

[AUTH-01] Q: How do I log in to CareBridge?

Intent: Informational — authentication & security

Answer: Call POST /auth/login with your registered email and password. On success the API returns a JWT access token valid for 24 hours. That token must be sent as an 'Authorization: Bearer {token}' header on every subsequent protected request; CareBridge uses stateless JWT authentication, so no server-side session is stored.

Alternative phrasings: just wondering, how do I log in to CareBridge?; How do I log in to CaerBridge?; hey carebot, how do i log in to carebridge; login?

Possible follow-ups: What happens when my token expires? | Is my password stored safely?

Related topics: AUTH-02, AUTH-03, SEC-01 **Keywords:** login, sign in, jwt token **Persona:** Donor **Difficulty:** Basic

[AUTH-02] Q: Why do I keep getting logged out / getting a 401 error?

Intent: Informational — authentication & security

Answer: A 401 Unauthorized response on a protected endpoint almost always means your JWT has expired. Tokens are valid for 24 hours from login; once that window passes you must log in again via POST /auth/login to obtain a fresh token.

Alternative phrasings: so why do I keep getting logged out / getting a 401 error?; Why do I keep getting logged out / getting a 401 error?; hey carebot, why do i keep getting logged out / getting a 401 error; 401 error?

Possible follow-ups: Can the 24-hour expiry be extended? | Do I lose my data when my session expires?

Related topics: AUTH-01, TRBL-02 **Keywords:** 401 error, logged out, token expired **Persona:** Donor **Difficulty:** Basic

[AUTH-03] Q: I forgot my password — how do I reset it?

Intent: Informational — authentication & security

Answer: Go to the forgot-password flow: POST /auth/forgot-password sends an OTP to your registered email, POST /auth/verify-forgot-otp checks that OTP, and POST /auth/reset-password lets you set a new password. Afterward, log in normally with the new password.

Alternative phrasings: hey, i forgot my password — how do I reset it?; I forgot my passowrd — how do I reset it?; hey carebot, i forgot my password — how do i reset it; forgot password?

Possible follow-ups: How long is the reset OTP valid? | What if I never receive the reset email?

Related topics: AUTH-01, OTP-01, TRBL-03 **Keywords:** forgot password, reset password, cant login **Persona:** Donor **Difficulty:** Basic

[SEC-01] Q: How are passwords and roles secured on the backend?

Intent: Informational — authentication & security

Answer: Passwords are hashed with BCrypt and never stored in plain text. Role-based access control restricts endpoints by ROLE_DONOR, ROLE_ORPHANAGE, and ROLE_ADMIN in SecurityConfig, and a JwtAuthFilter intercepts every request, validates the token's signature and expiry, and populates the SecurityContext. Session policy is STATELESS — every request must carry a valid JWT rather than relying on a server-side session.

Alternative phrasings: so how are passwords and roles secured on the backend?; How are passwords adn roles secured on the backend?; hey carebot, how are passwords and roles secured on the backend; security architecture?

Possible follow-ups: Which endpoints are public and require no authentication? | What security hardening is still planned but not yet implemented?

Related topics: AUTH-01, SEC-02, TECH-06 **Keywords:** security architecture, bcrypt, rbac, jwt filter **Persona:** Developer **Difficulty:** Advanced

[SEC-02] Q: Which API endpoints can be accessed without logging in?

Intent: Informational — authentication & security

Answer: SecurityConfig permits unauthenticated access to /auth/** (registration, login, verification), /api/landing/** (public landing page data), and the API documentation routes /swagger-ui/** and /v3/api-docs/**. Every other route requires a valid JWT and the correct role.

Alternative phrasings: so which API endpoints can be accessed without logging in?; Which API endpoints can be accessed without logging in?; hey carebot, which api endpoints can be accessed without logging in; public endpoints?

Possible follow-ups: What data is exposed on the public landing endpoints? | How is CORS configured for these public routes?

Related topics: SEC-01, LAND-01 **Keywords:** public endpoints, no auth required, open routes **Persona:** Developer

Difficulty: Advanced

5.x Donor Profile

[PROFILE-01] Q: What information do I need to create my donor profile?

Intent: Informational — donor profile

Answer: POST /donor/create-profile (multipart/form-data) requires your name, date of birth, designation, phone number, PAN number, and a PAN photo upload. Optional fields include a profile picture and address details (house number, mandal, village, district, state, country). The system then generates a unique CareBridgeID for your profile and sets its status to PENDING.

Alternative phrasings: just wondering, what information do I need to create my donor profile?; What information do I need to create my donor profile?; hey carebot, what information do i need to create my donor profile; donor profile fields?

Possible follow-ups: What format must my PAN number be in? | Can I update these details after submitting?

Related topics: PROFILE-02, PROFILE-03, VERIF-01 **Keywords:** donor profile fields, what documents needed, create profile

Persona: Donor **Difficulty:** Basic

[PROFILE-02] Q: What format does the PAN number need to follow?

Intent: Informational — donor profile

Answer: The PAN number is validated against the standard Indian pattern of five letters, four digits, and one letter (e.g., AAAAA9999A). It must also be unique — the system rejects a PAN number already tied to another profile.

Alternative phrasings: so what format does the PAN number need to follow?; What format does the PAN number need to follow?; hey carebot, what format does the pan number need to follow; pan format?

Possible follow-ups: What happens if my PAN is already registered by someone else? | Is the PAN photo required as well as the number?

Related topics: PROFILE-01, VERIF-04 **Keywords:** pan format, pan validation, pan pattern **Persona:** Donor **Difficulty:** Basic

[PROFILE-03] Q: Can I update my donor profile after it's been submitted?

Intent: Informational — donor profile

Answer: Yes. PUT /donor/update-profile (multipart/form-data) lets you update fields such as designation, address, phone, pictures, and PAN. Updates are possible both before and after admin approval; simple field changes do not require re-approval, though sensitive fields like PAN or date of birth may warrant another review.

Alternative phrasings: quick q - can I update my donor profile after it's been submitted?; Can I update my donor profile after it's been submitted?; hey carebot, can i update my donor profile after it's been submitted; edit profile?

Possible follow-ups: Do I lose my VERIFIED status when I edit my profile? | Which fields can't be changed after creation?

Related topics: PROFILE-01, VERIF-01 **Keywords:** edit profile, update donor details, change profile **Persona:** Donor
Difficulty: Basic

[PROFILE-04] Q: What is a CareBridgeID and where do I get one?

Intent: Informational — donor profile

Answer: A CareBridgeID is a unique identifier auto-generated by the backend for every donor and orphanage profile. It is distinct from your internal user ID and is the reference used across donations, needs, and slot bookings. You don't choose it — the system assigns it automatically when your profile is created.

Alternative phrasings: just wondering, what is a CareBridgeID and where do I get one?; What is a CareBridgeID and where do I get one?; hey carebot, what is a carebridgeid and where do I get one; carebridge id?

Possible follow-ups: Is CareBridgeID the same as my login email? | Where is my CareBridgeID displayed?

Related topics: PROFILE-01, PROFILE-05 **Keywords:** carebridge id, what is carebridgeid, profile id **Persona:** Donor
Difficulty: Basic

5.x Orphanage Profile

[PROFILE-05] Q: What documents does an orphanage need to submit to register?

Intent: Informational — orphanage profile

Answer: POST /orphanage/create-profile requires the orphanage name, admin name, number of children, email, phone, district, state, country, a DARPAN ID (social-sector registration), a PAN number with PAN photo, and a JJ Act Certificate photo (proof of Juvenile Justice Act compliance). Village, mandal, website link, social media links, and profile pictures are optional.

Alternative phrasings: quick q - what documents does an orphanage need to submit to register?; What documents does an orphanage need to submit to register?; hey carebot, what documents does an orphanage need to submit to register; orphanage documents?

Possible follow-ups: What is a DARPAN ID and why is it required? | What happens if a document is unclear or invalid?

Related topics: REG-05, VERIF-02, VERIF-04 **Keywords:** orphanage documents, darpan, jj act certificate **Persona:** Orphanage Admin
Difficulty: Basic

[PROFILE-06] Q: What is DARPAN ID and why does CareBridge need it?

Intent: Informational — orphanage profile

Answer: DARPAN ID is the registration number issued under India's social-sector scheme, and CareBridge requires it as one of the mandatory documents proving an orphanage is a legitimate, registered organization before it can be verified and allowed to post needs or accept donations.

Alternative phrasings: just wondering, what is DARPAN ID and why does CareBridge need it?; What is DARPAN ID and why does CareBridge need it?; hey carebot, what is darpan id and why does carebridge need it; darpan meaning?

Possible follow-ups: Is DARPAN checked automatically or manually by admins? | What other documents are checked alongside DARPAN?

Related topics: PROFILE-05, VERIF-02 **Keywords:** darpan meaning, why darpan required **Persona:** Orphanage Admin
Difficulty: Intermediate

5.x Verification & Admin Approval

[VERIF-01] Q: Why is my profile still showing as pending?

Intent: Informational — verification & admin approval

Answer: PENDING means your profile has been created and submitted but has not yet been reviewed by a platform admin. Approval is a manual step: an admin views your documents via GET /admin/donor/pending (or the orphanage equivalent) and then approves or rejects the profile. If it has been pending an unusually long time, that typically means the admin queue simply hasn't reached it yet.

Alternative phrasings: so why is my profile still showing as pending?; Why is my profile still showing as pending?; hey carebot, why is my profile still showing as pending; pending status?

Possible follow-ups: How long does approval usually take? | What happens if my profile is rejected instead?

Related topics: VERIF-02, VERIF-03, TRBL-04 **Keywords:** pending status, profile not approved, waiting for approval

Persona: Donor **Difficulty:** Basic

[VERIF-02] Q: How does an admin approve or reject a donor or orphanage profile?

Intent: Informational — verification & admin approval

Answer: Admins call PATCH /admin/donor/approve/{id} or PATCH /admin/orphanage/approve/{id} to set status to VERIFIED and trigger an approval email. To reject, PATCH /admin/donor/reject/{id} or the orphanage equivalent accepts an EmailReason payload; this moves the profile into the RejectedDonorProfile or RejectedOrpProfile archive table, deletes the original profile record, and emails the reason to the applicant.

Alternative phrasings: hey, how does an admin approve or reject a donor or orphanage profile?; How does an admin approve or reject a donor or orphanage profile?; hey carebot, how does an admin approve or reject a donor or orphanage profile; admin approve?

Possible follow-ups: Can a rejected user resubmit the same profile? | Is there an audit trail of admin decisions?

Related topics: VERIF-01, VERIF-03, ADMIN-02 **Keywords:** admin approve, admin reject, verification workflow **Persona:**

Platform Admin **Difficulty:** Intermediate

[VERIF-03] Q: Why was my profile rejected?

Intent: Informational — verification & admin approval

Answer: Common rejection reasons include an invalid PAN format or a blurry PAN photo, a PAN mismatch against your stated name or date of birth, a PAN number already registered by someone else, invalid or unclear supporting documents, or generally suspicious or incomplete information. Rejections are permanent for that submission — you must register again with corrected details rather than editing the rejected record.

Alternative phrasings: quick q - why was my profile rejected?; Why was my profile rejected?; hey carebot, why was my profile rejected; rejected profile?

Possible follow-ups: Can I see the specific reason given for my rejection? | Do I need a new email address to re-register?

Related topics: VERIF-01, VERIF-02 **Keywords:** rejected profile, why rejected, profile denied **Persona:** Donor **Difficulty:** Basic

[VERIF-04] Q: What's the difference between email verification and profile verification?

Intent: Informational — verification & admin approval

Answer: Email verification simply confirms you own the email address you registered with, and happens during the OTP step of registration. Profile verification is a separate, later step: an admin reviews your submitted documents and sets your status to PENDING, VERIFIED, or REJECTED. You can be email-verified while your profile is still PENDING.

Alternative phrasings: just wondering, what's the difference between email verification and profile verification?; What's the difference between email verification and profile verification?; hey carebot, what's the difference between email verification and profile verification; email verification vs profile status?

Possible follow-ups: Can I browse needs while my profile is still PENDING? | Does profile verification ever expire?

Related topics: VERIF-01, REG-02 **Keywords:** email verification vs profile status, verification confusion **Persona:** New Visitor **Difficulty:** Basic

[VERIF-05] Q: Can I donate before my profile is verified?

Intent: Informational — verification & admin approval

Answer: No. The required sequence is: register and verify your email via OTP, create your full profile with PAN and personal details, and then wait for an admin to approve it (you'll receive a VERIFIED-status confirmation email). Only after that can you donate or book visits.

Alternative phrasings: so can I donate before my profile is verified?; Can I donate before my profile is verified?; hey carebot, can I donate before my profile is verified; donate before verification?

Possible follow-ups: What can I do on the platform while pending? | How will I know when I'm approved?

Related topics: VERIF-01, DON-01 **Keywords:** donate before verification, donate while pending **Persona:** Donor **Difficulty:** Basic

5.x Need Items & Categories

[NEED-01] Q: How do I post a need for my orphanage?

Intent: Informational — need items & categories

Answer: Call POST /orphanage/needItem/create-item with a name (e.g. 'School Uniforms'), a category, quantity, price per quantity, priority, and optionally a description and product links. This creates a NeedItem with a unique needItemId, isActive set to true, and fulfilledQuantity starting at zero.

Alternative phrasings: just wondering, how do I post a need for my orphanage?; How do I post a need for my orphanage?; hey carebot, how do I post a need for my orphanage; post a need?

Possible follow-ups: What categories can a need item belong to? | Can I edit a need item after donors start donating to it?

Related topics: NEED-02, NEED-03, DON-01 **Keywords:** post a need, create need item, add requirement **Persona:** Orphanage Admin **Difficulty:** Basic

[NEED-02] Q: What categories of needs can orphanages post?

Intent: Informational — need items & categories

Answer: The CategoryType enum covers FOOD, EDUCATION (books, school supplies, uniforms), HEALTH (medicines, medical equipment), CLOTHING, TOYS (recreational items), SPORTS (sports equipment), TECHNOLOGY (computers, tablets, internet access), and OTHER.

Alternative phrasings: hey, what categories of needs can orphanages post?; What categories of needs can orphanages post?; hey carebot, what categories of needs can orphanages post; need categories?

Possible follow-ups: Can I filter needs by category when browsing? | What does priority level tell me about a need?

Related topics: NEED-01, NEED-04, BROWSE-01 **Keywords:** need categories, donation categories, types of needs

Persona: Donor **Difficulty:** Basic

[NEED-03] Q: What do the priority levels LOW, MEDIUM, HIGH, and URGENT mean for a need?

Intent: Informational — need items & categories

Answer: Priority signals how critical a need is: LOW means 'nice to have', MEDIUM means 'important', HIGH means 'significant need', and URGENT means 'critical/immediate need'. These labels help donors decide which needs to prioritize and support impact messaging on the platform.

Alternative phrasings: just wondering, what do the priority levels LOW, MEDIUM, HIGH, and URGENT mean for a need?; What do the priority levels LOW, MEDIUM, HIGH, and URGENT mean for a need?; hey carebot, what do the priority levels low, medium, high, and urgent mean for a need; priority meaning?

Possible follow-ups: Are urgent needs highlighted anywhere for donors? | Who sets the priority — the orphanage or the platform?

Related topics: NEED-02, DASH-06 **Keywords:** priority meaning, urgent needs, need priority levels **Persona:** Donor
Difficulty: Basic

[NEED-04] Q: Can more than one donor contribute to the same need?

Intent: Informational — need items & categories

Answer: Yes. CareBridge tracks total quantity needed, fulfilledQuantity (already delivered), and reservedQuantity (pledged by donors who haven't delivered yet). Remaining quantity is calculated as total minus fulfilled minus reserved, so multiple donors can each cover a portion of the same need until it is fully met.

Alternative phrasings: quick q - can more than one donor contribute to the same need?; Can more than one donor contribute to the same need?; hey carebot, can more than one donor contribute to the same need; multiple donors same need?

Possible follow-ups: What happens once a need is fully fulfilled? | Can I see how much of a need is still remaining before I donate?

Related topics: NEED-01, DON-04 **Keywords:** multiple donors same need, shared donation, partial fulfillment **Persona:** Donor
Difficulty: Intermediate

[NEED-05] Q: How do I edit or remove a need item I posted?

Intent: Informational — need items & categories

Answer: PUT /orphanage/needItem/updateItem/{id} lets you modify name, quantity, price, priority, description, and product links (the needItemId and owning orphanage cannot be changed). DELETE /orphanage/needItem/deleteItem/{id} removes it — typically a soft delete that sets isActive to false, with a copy retained in the DeletedItems table for audit history.

Alternative phrasings: hey, how do I edit or remove a need item I posted?; How do I edit or remove a need item I posted?; hey carebot, how do I edit or remove a need item I posted; edit need?

Possible follow-ups: Is a deleted need item gone permanently? | Can I un-delete a need item?

Related topics: NEED-01, NEED-06 **Keywords:** edit need, delete need item, remove requirement **Persona:** Orphanage Admin
Difficulty: Basic

[NEED-06] Q: Where can I see all the needs my orphanage has posted?

Intent: Informational — need items & categories

Answer: GET /orphanage/needItem/activeNeeds lists all needs your orphanage has posted, along with fulfillment progress for each item so you can track how close every need is to being fully met.

Alternative phrasings: just wondering, where can I see all the needs my orphanage has posted?; Where can I see all the needs my orphanage has posted?; hey carebot, where can i see all the needs my orphanage has posted; view my needs?

Possible follow-ups: Can donors also see this same list? | How is fulfillment progress calculated?

Related topics: NEED-01, NEED-04 **Keywords:** view my needs, active needs list, orphanage needs page **Persona:** Orphanage Admin **Difficulty:** Basic

[NEED-07] Q: How is a unique needItemId generated?

Intent: Informational — need items & categories

Answer: A dedicated utility class, NeedItemIdGenerator, is responsible for producing each unique needItemId when a new NeedItem is created by NeedItemService.

Alternative phrasings: hey, how is a unique needItemId generated?; How is a unique needItemId generated?; hey carebot, how is a unique needitemid generated; needitemid generation?

Possible follow-ups: What format does the generated needItemId follow? | Is the CareBridgeID generated by a similar dedicated utility?

Related topics: NEED-01, PROFILE-04 **Keywords:** needitemid generation, unique id logic **Persona:** Developer **Difficulty:** Advanced

5.x Donations

[DON-01] Q: How do I make a donation on CareBridge?

Intent: Informational — donations

Answer: Call POST /donor/donate with the needItemId you're supporting, a quantity, a donationType (IN_PERSON or ONLINE_ORDER), and type-specific details (expectedVisitDateTime for in-person, or expectedDeliveryDate, platformName, trackingId, and an order proof image for online orders). The donation is created with status PENDING and returns a donationRequestId.

Alternative phrasings: so how do I make a donation on CareBridge?; How do I make a donation on CareBridge?; hey carebot, how do i make a donation on carebridge; how to donate?

Possible follow-ups: What's the difference between IN_PERSON and ONLINE_ORDER donations? | Can I cancel a donation after creating it?

Related topics: DON-02, DON-03, DON-05 **Keywords:** how to donate, make a donation, donate items **Persona:** Donor **Difficulty:** Basic

[DON-02] Q: What's the difference between an IN_PERSON donation and an ONLINE_ORDER donation?

Intent: Informational — donations

Answer: IN_PERSON donations mean you deliver the item yourself during a scheduled visit; you must book a slot through the visit-booking system and provide an expectedVisitDateTime. ONLINE_ORDER donations mean you purchase the item online (e.g., from Amazon) and have it shipped; you supply an expectedDeliveryDate, the platformName, a trackingId, and an orderProofImage instead of booking a slot.

Alternative phrasings: hey, what's the difference between an IN_PERSON donation and an ONLINE_ORDER donation?; What's the difference between an IN_PERSON donation and an ONLINE_ORDER donation?; hey carebot, what's the difference between an in_person donation and an online_order donation; in person vs online donation?

Possible follow-ups: Do online-order donations ever need a slot too? | Who verifies that an online order actually arrived?

Related topics: DON-01, SLOT-01, DON-06 **Keywords:** in person vs online donation, donation types **Persona:** Donor
Difficulty: Basic

[DON-03] Q: Can I cancel a donation I already made?

Intent: Informational — donations

Answer: Only while it is still PENDING — DELETE /donor/delete-pending-donation/{id} removes it and frees up the reserved quantity for other donors. Once a donation's status is DELIVERED it cannot be cancelled, because the item has already been received and confirmed by the orphanage.

Alternative phrasings: so can I cancel a donation I already made?; Can I cancel a donation I already made?; hey carebot, can I cancel a donation I already made; cancel donation?

Possible follow-ups: Does the orphanage get notified if I cancel? | Can I cancel after the orphanage has confirmed delivery?

Related topics: DON-01, DON-04 **Keywords:** cancel donation, delete donation, undo donation **Persona:** Donor **Difficulty:** Basic

[DON-04] Q: Can I edit a donation after I've submitted it?

Intent: Informational — donations

Answer: Yes, via PATCH /donor/update/{donationId}, but only while the donation is still PENDING. You can change quantity, donation type, visit date, delivery date, platform, tracking ID, message, and the order proof image. You cannot switch which need item the donation is tied to — that requires creating a new donation. Once DELIVERED, nothing can be modified.

Alternative phrasings: just wondering, can I edit a donation after I've submitted it?; Can I edit a donation after I've submitted it?; hey carebot, can I edit a donation after I've submitted it; update donation?

Possible follow-ups: What if I picked the wrong need item entirely? | Can I add a delivery photo after the fact?

Related topics: DON-01, DON-03 **Keywords:** update donation, edit donation details, change donation **Persona:** Donor
Difficulty: Basic

[DON-05] Q: Where can I see the status of my past donations?

Intent: Informational — donations

Answer: Four donor-side endpoints cover this: GET /donor/pending-donations for donations awaiting delivery, GET /donor/completed-donations and GET /donor/delivered-donations for delivered items, and GET /donor/cancelled-donations for cancelled ones.

Alternative phrasings: quick q - where can I see the status of my past donations?; Where can I see the status of my past donations?; hey carebot, where can I see the status of my past donations; donation history?

Possible follow-ups: What does each donation status actually mean? | Is there a single combined view of all my donations?

Related topics: DON-01, DASH-01 **Keywords:** donation history, track my donations, donation status **Persona:** Donor
Difficulty: Basic

[DON-06] Q: How does an orphanage confirm that a donation was delivered?

Intent: Informational — donations

Answer: POST /orphanage/delivered/{id} accepts an ItemDeliveredDTO (multipart) with an optional delivery note and a delivered-photo upload. This sets the donation's status to DELIVERED, records the delivery date, note, and photo URL, and triggers an email notification to the donor.

Alternative phrasings: quick q - how does an orphanage confirm that a donation was delivered?; How does an orphanage confirm that a donation was delivered?; hey carebot, how does an orphanage confirm that a donation was delivered; confirm delivery?

Possible follow-ups: What if a delivery attempt fails? | Does confirming delivery affect the need item's fulfilled quantity?

Related topics: DON-01, DON-07 **Keywords:** confirm delivery, mark donation delivered, orphanage receive item **Persona:** Orphanage Admin **Difficulty:** Basic

[DON-07] Q: What happens if a delivery to us fails or the items never show up?

Intent: Informational — donations

Answer: If a delivery fails, the orphanage can decline to confirm it; the donation then returns to PENDING status rather than being marked DELIVERED, so it still appears in the pending queue for follow-up or eventual cancellation by the donor.

Alternative phrasings: just wondering, what happens if a delivery to us fails or the items never show up?; What happens if a delivery to us fails or the items never show up?; hey carebot, what happens if a delivery to us fails or the items never show up; failed delivery?

Possible follow-ups: Can the orphanage message the donor about a failed delivery? | Who initiates cancelling a donation that never arrived?

Related topics: DON-06, DON-03 **Keywords:** failed delivery, item never arrived, delivery rejected **Persona:** Orphanage Admin **Difficulty:** Intermediate

[DON-08] Q: Does CareBridge process payments for donations?

Intent: Informational — donations

Answer: No. CareBridge is currently not a payment platform. Donations take the form of in-person item delivery or online-ordered item delivery; any actual payment (for example, paying Amazon for the item) is arranged separately by the donor outside the platform.

Alternative phrasings: so does CareBridge process payments for donations?; Does CareBridge process payments for donations?; hey carebot, does carebridge process payments for donations; payment methods?

Possible follow-ups: Is payment-gateway integration planned for the future? | How do I prove I paid for an online-order donation?

Related topics: DON-02, ROAD-02 **Keywords:** payment methods, does carebridge accept payments, pay online **Persona:** New Visitor **Difficulty:** Basic

[DON-09] Q: Are my donations anonymous to the orphanage?

Intent: Informational — donations

Answer: No. CareBridge is built around transparency and accountability, so the receiving orphanage can see your name and profile information, and you can see the orphanage's details in turn. Donors also receive impact notifications tied to their donations.

Alternative phrasings: quick q - are my donations anonymous to the orphanage?; Are my donations anonymous to the orphanage?; hey carebot, are my donations anonymous to the orphanage; anonymous donation?

Possible follow-ups: Is there any way to donate without revealing my identity? | What exactly does the orphanage see about me?

Related topics: DON-01, DASH-04 **Keywords:** anonymous donation, does orphanage see my name, donor privacy **Persona:** Donor **Difficulty:** Basic

[DON-10] Q: What are all the possible donation status values?

Intent: Informational — donations

Answer: The documented DonationStatus values are PENDING (created, awaiting delivery), DELIVERED (received and confirmed by the orphanage), and CANCELLED (cancelled by the donor or system). A COMPLETED status is noted as a possible alternate status for fully fulfilled donations, though its exact usage alongside DELIVERED is not fully detailed in the knowledge base.

Alternative phrasings: hey, what are all the possible donation status values?; What are all the possible donation status values?; hey carebot, what are all the possible donation status values; donation status enum?

Possible follow-ups: Is COMPLETED used differently from DELIVERED in practice? | Can a CANCELLED donation ever be reinstated?

Related topics: DON-01, GAP-01 **Keywords:** donation status enum, all donation states **Persona:** Developer **Difficulty:** Advanced

5.x Visit Slots & Booking

[SLOT-01] Q: How do I create visit slots for donors as an orphanage?

Intent: Informational — visit slots & booking

Answer: POST /orphanage/slot/create takes a slotDate, startTime, endTime, and capacity (maximum visitors), and returns a SlotResponse with the new slotId and status. You can create multiple slots across different dates and times; each is a one-off window rather than a recurring schedule.

Alternative phrasings: so how do I create visit slots for donors as an orphanage?; How do I create visit slots for donors as an orphanage?; hey carebot, how do i create visit slots for donors as an orphanage; create visit slot?

Possible follow-ups: Do slots expire automatically? | Can I cancel a slot after creating it?

Related topics: SLOT-02, SLOT-03, BOOK-01 **Keywords:** create visit slot, add availability, orphanage schedule **Persona:** Orphanage Admin **Difficulty:** Basic

[SLOT-02] Q: How do I find available visit slots for a specific orphanage?

Intent: Informational — visit slots & booking

Answer: Call GET /donor/slots/available-slots/{orphanageCareBridgeId} to list every AVAILABLE slot for that orphanage, including date, time, capacity, and remaining open spots.

Alternative phrasings: hey, how do I find available visit slots for a specific orphanage?; How do I find available visit slots for a specific orphanage?; hey carebot, how do i find available visit slots for a specific orphanage; browse slots?

Possible follow-ups: What does it mean if a slot shows as FULL? | How do I actually book one of these slots?

Related topics: SLOT-01, BOOK-01 **Keywords:** browse slots, available visit times, check orphanage schedule **Persona:** Donor **Difficulty:** Basic

[SLOT-03] Q: What do the slot statuses AVAILABLE, EXPIRED, CANCELLED, and FULL mean?

Intent: Informational — visit slots & booking

Answer: AVAILABLE means the slot can still be booked. EXPIRED means its scheduled time has already passed — this is updated automatically by SlotSchedulerService, which runs scheduled jobs to expire past slots and clean up stale bookings. CANCELLED means the orphanage withdrew the slot. FULL means every capacity spot has been reserved.

Alternative phrasings: so what do the slot statuses AVAILABLE, EXPIRED, CANCELLED, and FULL mean?; What do the slot statuses AVAILABLE, EXPIRED, CANCELLED, and FULL mean?; hey carebot, what do the slot statuses available, expired, cancelled, and full mean; slot status meaning?

Possible follow-ups: Can an expired slot be reopened? | Does the scheduler run in real time or on a fixed interval?

Related topics: SLOT-01, SLOT-02 **Keywords:** slot status meaning, expired slot, full slot **Persona:** Donor **Difficulty:** Intermediate

[BOOK-01] Q: How do I book a visit to an orphanage?

Intent: Informational — visit slots & booking

Answer: POST /donor/slots/book takes the slotId, the number of visitors, and an optional message. The system validates that the number of visitors doesn't exceed remaining capacity, the slot is still AVAILABLE, and your donor profile is VERIFIED. A successful booking is created with status PENDING and returns a bookingId.

Alternative phrasings: just wondering, how do I book a visit to an orphanage?; How do I book a visit to an orphanage?; hey carebot, how do I book a visit to an orphanage; book a visit?

Possible follow-ups: What happens after I book — do I need orphanage confirmation? | Why did my booking attempt get rejected?

Related topics: SLOT-02, BOOK-02, BOOK-03 **Keywords:** book a visit, schedule orphanage visit, reserve slot **Persona:** Donor **Difficulty:** Basic

[BOOK-02] Q: How do I know if my visit booking is confirmed?

Intent: Informational — visit slots & booking

Answer: After you book, the orphanage reviews it and calls PATCH /orphanage/slot/confirm/{bookingId} to set status to CONFIRMED, or PATCH /orphanage/slot/reject/{bookingId} with a reason to set it to REJECTED. You'll be notified of the decision either way.

Alternative phrasings: so how do I know if my visit booking is confirmed?; How do I know if my visit booking is confirmed?; hey carebot, how do I know if my visit booking is confirmed; visit confirmation?

Possible follow-ups: Can I change my visit booking after it's confirmed? | What reasons might an orphanage reject a visit?

Related topics: BOOK-01, BOOK-03 **Keywords:** visit confirmation, booking status, is my visit approved **Persona:** Donor **Difficulty:** Basic

[BOOK-03] Q: How does an orphanage manage incoming visit booking requests?

Intent: Informational — visit slots & booking

Answer: GET /orphanage/slot/pending lists every PENDING visit booking, showing donor details and visitor counts, so the orphanage can decide whether to confirm or reject each one. Notifications are sent to the donor once a decision is made.

Alternative phrasings: hey, how does an orphanage manage incoming visit booking requests?; How does an orphanage manage incoming visit booking requests?; hey carebot, how does an orphanage manage incoming visit booking requests; manage

bookings?

Possible follow-ups: Can an orphanage bulk-approve multiple bookings? | What data does the orphanage see about the visiting donor?

Related topics: BOOK-01, BOOK-02 **Keywords:** manage bookings, review visit requests, orphanage confirm visit **Persona:** Orphanage Admin **Difficulty:** Basic

[SLOT-04] Q: Can I reuse the same booked slot for more than one visit?

Intent: Informational — visit slots & booking

Answer: No — one slot booking corresponds to one visit opportunity. Multiple donations can occur during that single visit, but if you want to visit again you need to book a separate slot for a new date and time.

Alternative phrasings: hey, can I reuse the same booked slot for more than one visit?; Can I reuse teh same booked slot for more than one visit?; hey carebot, can i reuse the same booked slot for more than one visit; reuse slot?

Possible follow-ups: Can I donate multiple items during one visit? | Is there a limit on how many slots I can book?

Related topics: BOOK-01, SLOT-01 **Keywords:** reuse slot, multiple visits same slot **Persona:** Donor **Difficulty:** Intermediate

[SLOT-05] Q: What's the difference between a slot and a donation?

Intent: Informational — visit slots & booking

Answer: A slot is a time window an orphanage makes available for visits; a donation is the actual transfer of items or value that a donor initiates. They work together for in-person giving — you must book a slot before an IN_PERSON donation visit — but ONLINE_ORDER donations don't require a slot at all.

Alternative phrasings: quick q - what's the difference between a slot and a donation?; What's teh difference between a slot and a donation?; hey carebot, what's the difference between a slot and a donation; slot vs donation difference?

Possible follow-ups: So can I donate without ever booking a slot? | Does an online donation ever get combined with a slot visit?

Related topics: DON-02, SLOT-01 **Keywords:** slot vs donation difference, confused slot donation **Persona:** Donor **Difficulty:** Basic

[SLOT-06] Q: What automated jobs keep slot data accurate over time?

Intent: Informational — visit slots & booking

Answer: SlotSchedulerService runs scheduled (cron-style) tasks that mark slots EXPIRED once their scheduled time has passed, clean up cancelled or finished bookings, and update availability counts, so donors never see a stale slot as bookable.

Alternative phrasings: just wondering, what automated jobs keep slot data accurate over time?; What auotmated jobs keep slot data accurate over time?; hey carebot, what automated jobs keep slot data accurate over time; slot scheduler?

Possible follow-ups: How frequently does the scheduler run? | Does the scheduler send any notification when it expires a slot?

Related topics: SLOT-03, TECH-02 **Keywords:** slot scheduler, cron jobs, automatic slot expiry **Persona:** Developer **Difficulty:** Advanced

5.x Dashboards & Analytics

[DASH-01] Q: What does my donor dashboard show me?

Intent: Informational — dashboards & analytics

Answer: The donor dashboard combines a profile summary (verification and subscription status, member-since date), donation statistics (counts, total value, breakdown by category and status), an impact summary (orphanages helped, estimated lives impacted, fulfillment contribution), your most recent donations, upcoming confirmed bookings, and a list of urgent needs recommended to you.

Alternative phrasings: hey, what does my donor dashboard show me?; What does my donor dashboard show me?; hey carebot, what does my donor dashboard show me; donor dashboard features?

Possible follow-ups: How is 'lives impacted' actually estimated? | How often does the dashboard data refresh?

Related topics: DASH-02, DASH-03 **Keywords:** donor dashboard features, what can I see on my dashboard **Persona:** Donor **Difficulty:** Basic

[DASH-02] Q: How is my donor impact score or 'lives impacted' calculated?

Intent: Informational — dashboards & analytics

Answer: Impact is derived from delivered (not pending) donations: total donations delivered, their total monetary value, the number of distinct orphanages helped, and the categories of needs fulfilled feed into an estimated lives-impacted figure — for example, fifty delivered uniforms translates to roughly fifty children impacted.

Alternative phrasings: quick q - how is my donor impact score or 'lives impacted' calculated?; How is my donor impact score or 'lives impacted' calculated?; hey carebot, how is my donor impact score or 'lives impacted' calculated; impact calculation?

Possible follow-ups: Do pending donations count toward my impact at all? | Can I see impact broken down by orphanage?

Related topics: DASH-01, DON-05 **Keywords:** impact calculation, lives impacted meaning, donor impact metric **Persona:** Donor **Difficulty:** Intermediate

[DASH-03] Q: What does the orphanage dashboard display?

Intent: Informational — dashboards & analytics

Answer: It provides an overview (total needs posted, total donations received, total visitors, overall fulfillment percentage), a donation summary (counts, value, breakdown by category and by IN_PERSON vs ONLINE_ORDER, recent timeline, pending count), a needs summary (active needs, quantities needed vs. fulfilled, average fulfillment by category), a visit summary (confirmed, upcoming, and past visits, average visitors per visit), and an activity feed of recent donations, new needs, and upcoming confirmations.

Alternative phrasings: just wondering, what does the orphanage dashboard display?; What does the orphanage dashboard display?; hey carebot, what does the orphanage dashboard display; orphanage dashboard features?

Possible follow-ups: Can the orphanage export this data? | How is the fulfillment percentage calculated?

Related topics: DASH-01, NEED-06 **Keywords:** orphanage dashboard features, orphanage analytics **Persona:** Orphanage Admin **Difficulty:** Basic

[DASH-04] Q: Where do I see my upcoming scheduled visits?

Intent: Informational — dashboards & analytics

Answer: The donor dashboard's 'Upcoming Bookings' section lists your next scheduled visits with slot details (date, time, orphanage), confirmation status, and reminder notifications.

Alternative phrasings: quick q - where do I see my upcoming scheduled visits?; Where do I see my upcoming scheduled visits?; hey carebot, where do I see my upcoming scheduled visits; upcoming visits?

Possible follow-ups: Will I get a reminder notification before the visit? | Can I cancel an upcoming visit from this view?

Related topics: DASH-01, BOOK-02 **Keywords:** upcoming visits, my scheduled bookings **Persona:** Donor **Difficulty:** Basic

[DASH-05] Q: What can a platform admin see on their dashboard?

Intent: Informational — dashboards & analytics

Answer: The admin dashboard shows overall platform statistics, the current count of pending approvals, recent activity across the system, and system health metrics, alongside separate views for pending, approved, and rejected donor and orphanage profiles.

Alternative phrasings: hey, what can a platform admin see on their dashboard?; What can a platform admin see on their dashboard?; hey carebot, what can a platform admin see on their dashboard; admin dashboard?

Possible follow-ups: Can an admin see individual donor donation histories? | Is there an approval efficiency or average review time metric?

Related topics: VERIF-02, ADMIN-01 **Keywords:** admin dashboard, admin overview **Persona:** Platform Admin **Difficulty:** Intermediate

[DASH-06] Q: How does CareBridge recommend needs to me?

Intent: Informational — dashboards & analytics

Answer: The donor dashboard surfaces 'Urgent Needs' — high-priority needs from verified orphanages, prioritizing orphanages in greatest need and matching against criteria relevant to donor interests.

Alternative phrasings: just wondering, how does CareBridge recommend needs to me?; How does CareBridge recommend needs to me?; hey carebot, how does carebridge recommend needs to me; need recommendations?

Possible follow-ups: Can I customize what kinds of needs get recommended to me? | Is this the same list as the general needs browsing page?

Related topics: NEED-03, DASH-01 **Keywords:** need recommendations, urgent needs recommendation **Persona:** Donor **Difficulty:** Basic

5.x Notifications & OTP

[OTP-01] Q: What is an OTP used for on CareBridge?

Intent: Informational — notifications & otp

Answer: OTPs (One-Time Passwords) are six-digit codes emailed for registration verification, password resets, and orphanage verification. Each OTP is stored with a five-minute expiry and a used flag, and becomes invalid once used or expired.

Alternative phrasings: just wondering, what is an OTP used for on CareBridge?; What is an OTP used for on CareBridge?; hey carebot, what is an otp used for on carebridge; what is otp?

Possible follow-ups: What if I never receive the OTP email at all? | How many times can I request a new OTP?

Related topics: REG-04, OTP-02, TRBL-05 **Keywords:** what is otp, one time password purpose **Persona:** New Visitor **Difficulty:** Basic

[OTP-02] Q: I never received my OTP email — what should I do?

Intent: Informational — notifications & otp

Answer: First check your spam or junk folder, since automated verification emails are sometimes filtered there. If it's still missing, use the resend endpoint to generate a new OTP (allowed roughly three to five times before a cooldown applies), and confirm the email address you entered was correct.

Alternative phrasings: hey, i never received my OTP email — what should I do?; I never received my OTP email — what should I do?; hey carebot, i never received my otp email — what should i do; otp not received?

Possible follow-ups: Is there a limit on resend attempts? | Does CareBridge use different email providers for dev vs. production?

Related topics: OTP-01, TRBL-05 **Keywords:** otp not received, missing verification email **Persona:** New Visitor **Difficulty:** Basic

[NOTIF-01] Q: What kinds of emails will CareBridge send me?

Intent: Informational — notifications & otp

Answer: Email notifications are triggered for: registration OTPs, password-reset OTPs, orphanage-registration OTPs, profile approval, profile rejection (with reason), donation creation confirmation, donation delivery confirmation, and visit booking confirmation or rejection.

Alternative phrasings: hey, what kinds of emails will CareBridge send me?; What kinds of emails will CareBridge send me?; hey carebot, what kinds of emails will carebridge send me; what emails will I get?

Possible follow-ups: Can I opt out of any of these notifications? | Are notifications also shown inside the app, not just by email?

Related topics: OTP-01, DON-06, BOOK-02 **Keywords:** what emails will I get, notification types **Persona:** Donor **Difficulty:** Basic

[NOTIF-02] Q: What email infrastructure does CareBridge use to send notifications?

Intent: Informational — notifications & otp

Answer: Development environments send mail through Gmail SMTP, while production uses the Brevo SMTP API, both configured through Spring Mail properties in application-dev.properties and application-prod.properties respectively. EmailMessages.java acts as a template library for the formatted content of each notification type.

Alternative phrasings: quick q - what email infrastructure does CareBridge use to send notifications?; What email infrastructure does CareBridge use to send notifications?; hey carebot, what email infrastructure does carebridge use to send notifications; smtp configuration?

Possible follow-ups: Does Cloudinary play any role in sending emails? | Where would I add a new email template?

Related topics: NOTIF-01, TECH-03 **Keywords:** smtp configuration, email service backend, brevo gmail **Persona:** Developer **Difficulty:** Advanced

5.x Messaging

[MSG-01] Q: Can I message an orphanage directly through CareBridge?

Intent: Informational — messaging

Answer: Yes, messaging components exist on both sides of the platform — DonorMessages.jsx and OrphanageMessagePage.jsx on the frontend, backed by a shared Messages.jsx component — allowing donors and orphanages to communicate within the platform.

Alternative phrasings: so can I message an orphanage directly through CareBridge?; Can I message an orphanage directly through CareBridge?; hey carebot, can i message an orphanage directly through carebridge; messaging feature?

Possible follow-ups: Is messaging real-time or does it require a page refresh? | Can admins see messages between donors and orphanages?

Related topics: DASH-01, ROAD-01 **Keywords:** messaging feature, chat with orphanage, contact orphanage directly
Persona: Donor **Difficulty:** Basic

5.x Landing Page & Browse

[LAND-01] Q: Can I see CareBridge's impact statistics without creating an account?

Intent: Informational — landing page & browse

Answer: Yes. GET /api/landing/stats is a public endpoint (no authentication needed) that returns global statistics such as total verified orphanages, total active donors, total donations received, total items distributed, and success-story counts, so visitors can see the platform's impact before registering.

Alternative phrasings: hey, can I see CareBridge's impact statistics without creating an account?; Can I see CaerBridge's impact statistics without creating an account?; hey carebot, can i see carebridge's impact statistics without creating an account; public stats?

Possible follow-ups: Can I browse specific orphanages before signing up? | Are these numbers updated in real time?

Related topics: SEC-02, LAND-02 **Keywords:** public stats, landing page numbers, impact without account **Persona:** New Visitor **Difficulty:** Basic

[LAND-02] Q: Can I browse orphanages and needs before registering?

Intent: Informational — landing page & browse

Answer: Yes. GET /api/landing/orphanages lists verified orphanages (name, photo, location, number of children, active needs, total donations, recent activity), and GET /api/landing/needs lists active needs — both are public endpoints intended for discovery and marketing before you create an account.

Alternative phrasings: hey, can I browse orphanages and needs before registering?; Can I browse orphanages adn needs before registering?; hey carebot, can i browse orphanages and needs before registering; browse before signup?

Possible follow-ups: Do I need an account to actually donate to something I see here? | Is this the same data donors see after logging in?

Related topics: LAND-01, VERIF-05 **Keywords:** browse before signup, view needs without login **Persona:** New Visitor **Difficulty:** Basic

5.x Admin Operations

[ADMIN-01] Q: What can a platform admin actually do on CareBridge?

Intent: Informational — admin operations

Answer: Admins can approve or reject donor and orphanage profiles, view overall platform statistics, monitor donations and visit bookings, manage system settings, view user data for verification purposes, and send notifications. Admins do not create donations as a donor, post needs as an orphanage, or book visits themselves.

Alternative phrasings: quick q - what can a platform admin actually do on CareBridge?; What can a pltaform admin actually do on CareBridge?; hey carebot, what can a platform admin actually do on carebridge; admin capabilities?

Possible follow-ups: Can an admin edit a donor's profile directly? | Is there more than one admin account typically?

Related topics: VERIF-02, DASH-05 **Keywords:** admin capabilities, what can admin do **Persona:** Platform Admin
Difficulty: Intermediate

[ADMIN-02] Q: What criteria do admins use when reviewing a profile?

Intent: Informational — admin operations

Answer: For donors, admins check for a valid PAN number format, a clear PAN photo, valid contact information, and no duplicate records. For orphanages, they check valid DARPAN registration, a valid JJ Act Certificate, a valid PAN number, legitimate contact information, and a verifiable registered address.

Alternative phrasings: hey, what criteria do admins use when reviewing a profile?; What criteria do admins use when reviewing a profile?; hey carebot, what criteria do admins use when reviewing a profile; review criteria?

Possible follow-ups: Are these criteria enforced automatically or manually? | What happens if only one document is invalid but the rest are fine?

Related topics: VERIF-02, PROFILE-01, PROFILE-05 **Keywords:** review criteria, what admins check, approval checklist
Persona: Platform Admin **Difficulty:** Intermediate

5.x Technical Architecture

[TECH-01] Q: What is the overall technology stack behind CareBridge?

Intent: Informational — technical architecture

Answer: The backend is Spring Boot 4.0.6 on Java 21 with Maven, using MySQL in production and H2 for testing, following a layered Controller-Service-Repository-Entity architecture with Hibernate as the ORM and Swagger/OpenAPI documentation. The donor/orphanage frontend is React 19.2.5 with Vite 8.0.10 and Bootstrap 5.3.8; the admin portal is a separate React 19.2.6 + Vite 8.0.12 application.

Alternative phrasings: so what is the overall technology stack behind CareBridge?; What is the overall technology stack behind CareBridge?; hey carebot, what is the overall technology stack behind carebridge; tech stack?

Possible follow-ups: Why are there three separate codebases instead of one? | What testing tools are used across the stack?

Related topics: TECH-02, TECH-04 **Keywords:** tech stack, what is carebridge built with **Persona:** Developer **Difficulty:** Advanced

[TECH-02] Q: How is the backend organized into modules?

Intent: Informational — technical architecture

Answer: The backend is split into twelve functional modules: authentication, donor profile management, orphanage profile management, donation management, visit-booking management, needs management, dashboard management (donor and orphanage), notification management, admin management, slot-details fetch management, landing-page data management, and common utilities (Cloudinary config, image upload, the shared ApiResponse wrapper).

Alternative phrasings: just wondering, how is the backend organized into modules?; How is the backend organized into modules?; hey carebot, how is the backend organized into modules; backend modules?

Possible follow-ups: Which module handles slot expiry automation? | Where do image uploads get processed?

Related topics: TECH-01, TECH-05 **Keywords:** backend modules, codebase structure, package organization **Persona:** Developer **Difficulty:** Advanced

[TECH-03] Q: How does CareBridge handle image and file uploads?*Intent:* Informational — technical architecture

Answer: ImageUploadService routes all uploads (profile pictures, PAN photos, certificates, donation and delivery photos) to Cloudinary via CloudinaryConfig, supporting JPG, PNG, and GIF up to 20MB per file and 100MB per request. Only the resulting Cloudinary URL is stored in the database — images themselves live in Cloudinary's CDN and are not deleted even if the related profile is later removed.

Alternative phrasings: just wondering, how does CareBridge handle image and file uploads?; How does CareBridge handle image and file uploads?; hey carebot, how does carebridge handle image and file uploads; file upload?

Possible follow-ups: What happens to an uploaded image if I delete my profile? | What upload size limit applies to a single request versus a single file?

Related topics: TECH-01, PROFILE-01 **Keywords:** file upload, cloudinary, image storage **Persona:** Developer **Difficulty:** Advanced

[TECH-04] Q: What are the core database entities and how do they relate?*Intent:* Informational — technical architecture

Answer: User relates 1:1 to DonorProfile and 1:1 to OrphanageProfile. OrphanageProfile relates 1:many to NeedItem and 1:many to Slot. DonorProfile relates 1:many to DonationRequest and 1:many to VisitBooking. NeedItem relates 1:many to DonationRequest, and Slot relates 1:many to VisitBooking. Archive tables (RejectedDonorProfile, RejectedOrpProfile, DeletedItems) preserve history when records are rejected or removed.

Alternative phrasings: so what are the core database entities and how do they relate?; What are the core database entities and how do they relate?; hey carebot, what are the core database entities and how do they relate; database schema?

Possible follow-ups: Why keep archive tables instead of just deleting records? | Is there a unique constraint on phone numbers across profiles?

Related topics: TECH-02, TECH-05 **Keywords:** database schema, entity relationships, erd **Persona:** Developer **Difficulty:** Advanced

[TECH-05] Q: How is data archived when a profile is rejected or a need item is deleted?*Intent:* Informational — technical architecture

Answer: Rejected donor and orphanage profiles are copied into RejectedDonorProfile and RejectedOrpProfile respectively (with the rejection reason attached) before the original record is deleted. Deleted need items are copied into the DeletedItems table with deletedBy, deletionReason, and deletedAt fields, preserving an audit trail rather than silently discarding data.

Alternative phrasings: just wondering, how is data archived when a profile is rejected or a need item is deleted?; How is data archived when a profile is rejected or a need item is deleted?; hey carebot, how is data archived when a profile is rejected or a need item is deleted; data archival?

Possible follow-ups: Can archived records ever be restored to active status? | Who can view these archive tables?

Related topics: TECH-04, VERIF-02 **Keywords:** data archival, audit trail, rejected profile storage **Persona:** Developer **Difficulty:** Advanced

[TECH-06] Q: What security practices are already implemented versus still planned?*Intent:* Informational — technical architecture

Answer: Implemented: BCrypt password hashing, stateless JWT authentication, role-based access control, CORS configuration, @Valid input validation, exception handling with error masking, multipart upload validation, OTP expiry/single-use enforcement, and ORM-based SQL-injection prevention. Still to implement: enforced HTTPS/SSL, API rate limiting, request logging and monitoring, two-factor authentication, audit logging for sensitive operations, encryption at rest, GDPR-style data deletion, and additional security headers such as X-Frame-Options and CSP.

Alternative phrasings: hey, what security practices are already implemented versus still planned?; What security practices are already implemented versus still planned?; hey carebot, what security practices are already implemented versus still planned; security roadmap?

Possible follow-ups: Which of the planned items would you prioritize first? | Is rate limiting needed before or after a payment gateway is added?

Related topics: SEC-01, GAP-01 **Keywords:** security roadmap, implemented vs planned security **Persona:** Developer
Difficulty: Advanced

[TECH-07] Q: How do I run CareBridge locally for development?

Intent: Informational — technical architecture

Answer: Backend: install a Java 21 SDK and Maven, run a local MySQL server, then run 'mvn spring-boot:run'. Frontend: install Node.js 18+, then run 'npm run dev' to start the Vite dev server (typically on port 5173). The admin portal follows the same Node.js/npm pattern as a separate app.

Alternative phrasings: quick q - how do I run CareBridge locally for development?; How do I run CareBridge locally for development?; hey carebot, how do I run carebridge locally for development; local setup?

Possible follow-ups: What database credentials does the backend expect locally? | How is the admin portal deployed differently from the main frontend?

Related topics: TECH-01, DEPLOY-01 **Keywords:** local setup, run project locally, dev environment **Persona:** Developer
Difficulty: Intermediate

[DEPLOY-01] Q: How is CareBridge deployed to production?

Intent: Informational — technical architecture

Answer: The backend can be deployed to any cloud host supporting a Java 21 runtime (AWS, Azure, Heroku, DigitalOcean) with a MySQL database and environment variables for database credentials, the JWT secret, SMTP credentials, Cloudinary keys, and CORS origins, built via 'mvn clean install' and run as a jar. The frontend and admin portal are each built with 'npm run build' and deployed independently to platforms like Firebase Hosting, Vercel, or Netlify.

Alternative phrasings: hey, how is CareBridge deployed to production?; How is CareBridge deployed to production?; hey carebot, how is carebridge deployed to production; deployment process?

Possible follow-ups: Are secrets like the JWT key stored in the codebase? | Can the frontend and admin portal be hosted on different domains?

Related topics: TECH-07, TECH-06 **Keywords:** deployment process, how is it hosted, production setup **Persona:** Developer
Difficulty: Intermediate

[API-01] Q: What does the standard CareBridge API response look like?

Intent: Informational — technical architecture

Answer: ApiResponse.java defines a consistent wrapper used across endpoints, with a boolean 'success' field, a 'message' string, and a 'data' object carrying the actual payload, ensuring every response follows the same shape regardless of which module handled the request.

Alternative phrasings: hey, what does the standard CareBridge API response look like?; What does teh standard CareBridge API response look like?; hey carebot, what does the standard carebridge api response look like; api response format?

Possible follow-ups: Do error responses use a different wrapper, like ErrorResponse? | Is this format documented in the Swagger spec?

Related topics: TECH-02, SEC-02 **Keywords:** api response format, apiresponse wrapper **Persona:** Developer **Difficulty:** Advanced

[API-02] Q: Where can I find the full, always-current CareBridge API specification?

Intent: Informational — technical architecture

Answer: The backend auto-generates OpenAPI documentation via springdoc-openapi-starter-webmvc-ui, browsable at /swagger-ui/** and machine-readable at /v3/api-docs/**, both public routes that don't require authentication.

Alternative phrasings: so where can I find the full, always-current CareBridge API specification?; Where can I find teh full, always-current CareBridge API specification?; hey carebot, where can i find the full, always-current carebridge api specification; api spec?

Possible follow-ups: Is the deployed Swagger UI publicly linked from the main site? | Does the OpenAPI spec include example request/response payloads?

Related topics: SEC-02, TECH-08 **Keywords:** api spec, swagger docs, openapi **Persona:** Developer **Difficulty:** Advanced

[RAG-DOC] Q: What AI stack powers the CareBot chatbot's answers?

Intent: Informational — technical architecture

Answer: CareBot uses Retrieval-Augmented Generation: an OpenRouter-hosted LLM generates the reply text, grounded by passages retrieved from a Qdrant Cloud vector database that stores embeddings of the platform's knowledge base (see Chapter 8 for chunking and retrieval guidance).

Alternative phrasings: so what AI stack powers the CareBot chatbot's answers?; What AI stack powers teh CareBot chatbot's answers?; hey carebot, what ai stack powers the carebot chatbot's answers; carebot ai stack?

Possible follow-ups: Which embedding model populates the Qdrant collection? | How is retrieval quality evaluated over time?

Related topics: FE-04, TECH-01 **Keywords:** carebot ai stack, rag architecture, qdrant openrouter **Persona:** Developer **Difficulty:** Advanced

5.x Troubleshooting

[TRBL-01] Q: I'm getting a CORS error in the browser — what does that mean?

Intent: Informational — troubleshooting

Answer: A CORS error ('Access to XMLHttpRequest blocked by CORS policy') means the frontend's origin isn't in the backend's allowed-origins list. The fix is on the backend configuration side: confirm the frontend URL is whitelisted in the CORS settings.

Alternative phrasings: just wondering, i'm getting a CORS error in the browser — what does that mean?; I'm getting a CORS error in teh browser — what does that mean?; hey carebot, i'm getting a cors error in the browser — what does that mean; cors error?

Possible follow-ups: What origins are allowed by default? | Does this differ between local development and production?

Related topics: TECH-06, TRBL-02 **Keywords:** cors error, blocked by cors policy **Persona:** New Visitor **Difficulty:** Basic

[TRBL-02] Q: Why do I suddenly get 'Unauthorized' errors while using the app?

Intent: Informational — troubleshooting

Answer: This is the standard symptom of an expired JWT (tokens last 24 hours from login). Logging in again via POST /auth/login issues a fresh token and resolves it.

Alternative phrasings: quick q - why do I suddenly get 'Unauthorized' errors while using the app?; Why do I suddenly get 'Unauthorized' errors while using teh app?; hey carebot, why do i suddenly get 'unauthorized' errors while using the app; unauthorized error?

Possible follow-ups: Is there a way to refresh a token without a full re-login? | Will I lose unsaved form data when this happens?

Related topics: AUTH-02 **Keywords:** unauthorized error, 401 while using app **Persona:** Donor **Difficulty:** Basic

[TRBL-03] Q: My file upload keeps failing with a 413 error — why?

Intent: Informational — troubleshooting

Answer: A 413 'Payload Too Large' response means the file exceeds the 20MB per-file limit, or the overall multipart request exceeds the 100MB total limit configured on the backend. Compress or resize the file and try again.

Alternative phrasings: quick q - my file upload keeps failing with a 413 error — why?; My file upolad keeps failing with a 413 error — why?; hey carebot, my file upload keeps failing with a 413 error — why; 413 error?

Possible follow-ups: What file formats are accepted for uploads? | Is there a recommended way to compress a PAN photo before upload?

Related topics: TECH-03 **Keywords:** 413 error, file too large, upload fails **Persona:** Donor **Difficulty:** Basic

[TRBL-04] Q: My profile has been PENDING for a long time — is something broken?

Intent: Informational — troubleshooting

Answer: Not necessarily. PENDING simply means an admin hasn't reviewed your profile yet; there's no automatic timeout. If it's been an unusually long wait, the practical next step is to check in with the platform admin directly.

Alternative phrasings: just wondering, my profile has been PENDING for a long time — is something broken?; My profile has been PENDING for a long time — is something broken?; hey carebot, my profile has been pending for a long time — is something broken; stuck pending?

Possible follow-ups: Is there an expected turnaround time for approvals? | Can I contact support about a delayed review?

Related topics: VERIF-01 **Keywords:** stuck pending, approval taking too long **Persona:** Donor **Difficulty:** Basic

[TRBL-05] Q: I can't create a donation — it says the orphanage wasn't found or quantity is insufficient. What's wrong?

Intent: Informational — troubleshooting

Answer: Double-check that the needItemId you're referencing is correct, that the need's remaining quantity is not already zero, and that the target orphanage's profile is VERIFIED — donations against an unverified or not-found orphanage/need will be rejected by the backend.

Alternative phrasings: hey, i can't create a donation — it says the orphanage wasn't found or quantity is insufficient. What's wrong?; I can't create a doantion — it says the orphanage wasn't found or quantity is insufficient. What's wrong?; hey carebot, i can't create a donation — it says the orphanage wasn't found or quantity is insufficient. what's wrong; cannot create donation error?

Possible follow-ups: How can I see the remaining quantity before I try to donate? | What if the need was deleted after I started the donation form?

Related topics: NEED-04, DON-01 **Keywords:** cannot create donation error, orphanage not found, insufficient quantity
Persona: Donor **Difficulty:** Basic

[TRBL-06] Q: Why does booking an available slot keep failing for me?

Intent: Informational — troubleshooting

Answer: Slot booking fails if your donor profile isn't yet VERIFIED, if the requested capacity has already been used up by other bookings, or if the slot's date/time has already passed. Confirm your verification status first, since that's the most common blocker.

Alternative phrasings: so why does booking an available slot keep failing for me?; Why does bokoing an available slot keep failing for me?; hey carebot, why does booking an available slot keep failing for me; slot booking fails?

Possible follow-ups: How do I check whether my donor profile is verified? | Can I get notified when a full slot opens back up?

Related topics: BOOK-01, VERIF-05 **Keywords:** slot booking fails, cant book visit **Persona:** Donor **Difficulty:** Basic

5.x Policies & Trust

[POL-01] Q: How does CareBridge make sure orphanages are legitimate?

Intent: Informational — policies & trust

Answer: Every orphanage must submit a DARPAN registration ID, a PAN number with photo, and a JJ Act Certificate proving Juvenile Justice Act compliance. A platform admin manually reviews these documents before granting VERIFIED status, and only verified orphanages can post needs or receive donations.

Alternative phrasings: quick q - how does CareBridge make sure orphanages are legitimate?; How does CaerBridge make sure orphanages are legitimate?; hey carebot, how does carebridge make sure orphanages are legitimate; how orphanages are verified?

Possible follow-ups: What happens if an orphanage's documents turn out to be fraudulent later? | Are verified orphanages ever re-reviewed periodically?

Related topics: PROFILE-05, VERIF-02 **Keywords:** how orphanages are verified, trust and safety, legitimacy check
Persona: New Visitor **Difficulty:** Basic

[POL-02] Q: How does CareBridge verify that donors are real people?

Intent: Informational — policies & trust

Answer: Donors must provide a valid PAN number in the standard Indian format along with a PAN photo, plus personal details like date of birth, designation, and a unique phone number. An admin reviews this before granting VERIFIED status, which is required before the donor can donate or book visits.

Alternative phrasings: so how does CareBridge verify that donors are real people?; How does CaerBridge verify that donors are real people?; hey carebot, how does carebridge verify that donors are real people; donor verification?

Possible follow-ups: Is any government ID besides PAN accepted? | Can a donor be re-verified if their PAN changes?

Related topics: PROFILE-01, VERIF-02 **Keywords:** donor verification, how donors are checked **Persona:** New Visitor
Difficulty: Basic

5.x Testing & Quality Assurance

[QA-01] Q: What testing tools and commands does the CareBridge backend use?

Intent: Informational — testing & quality assurance

Answer: Backend tests use Mockito for unit-level mocking and an H2 in-memory database for integration and repository tests, covering repository, service, and controller layers. Run the suite with 'mvn test', or 'mvn verify' for a full verification build.

Alternative phrasings: quick q - what testing tools and commands does the CareBridge backend use?; What testing tools and commands does the CareBridge backend use?; hey carebot, what testing tools and commands does the carebridge backend use; backend testing tools?

Possible follow-ups: Is there a documented code coverage target? | Do controller tests hit the real security filter chain?

Related topics: QA-02, TECH-01 **Keywords:** backend testing tools, mvn test, mockito h2 **Persona:** QA Engineer
Difficulty: Advanced

[QA-02] Q: How is the frontend tested?

Intent: Informational — testing & quality assurance

Answer: Component-level testing uses React Testing Library, while end-to-end flows are tested with Playwright or Cypress. Manual cross-browser verification is also performed across Chrome, Firefox, and Safari. ESLint enforces code quality, and SonarQube is listed as an optional static-analysis addition.

Alternative phrasings: so how is the frontend tested?; How is the frontend tested?; hey carebot, how is the frontend tested; frontend testing?

Possible follow-ups: Which flows have end-to-end coverage today? | Is there a CI pipeline that runs these tests automatically?

Related topics: QA-01, TECH-07 **Keywords:** frontend testing, react testing library, cypress playwright **Persona:** QA Engineer
Difficulty: Advanced

[QA-03] Q: What tool checks code quality on the frontend?

Intent: Informational — testing & quality assurance

Answer: ESLint is used for frontend code quality; SonarQube is mentioned as an optional tool for deeper code analysis, alongside JUnit for backend testing.

Alternative phrasings: just wondering, what tool checks code quality on the frontend?; What tool checks code quality on the frontend?; hey carebot, what tool checks code quality on the frontend; eslint?

Possible follow-ups: Are there custom ESLint rules beyond the defaults? | Is SonarQube currently wired into the build?

Related topics: QA-01, QA-02 **Keywords:** eslint, sonarqube, code quality tools **Persona:** Developer **Difficulty:** Advanced

5.x Performance & Scalability

[PERF-01] Q: What performance optimizations does CareBridge already have in place?

Intent: Informational — performance & scalability

Answer: Stateless JWT authentication avoids server-side session storage, frequently accessed fields (email, phone, CareBridgeID) are indexed, frontend components are lazy-loaded, images are optimized and CDN-delivered via Cloudinary, and Spring Data JPA manages database connection pooling.

Alternative phrasings: hey, what performance optimizations does CareBridge already have in place?; What performance optimizations does CareBridge already have in place?; hey carebot, what performance optimizations does carebridge already have in place; current optimizations?

Possible follow-ups: What's the biggest performance risk as usage grows? | Is there load-testing data for the current setup?

Related topics: PERF-02, TECH-01 **Keywords:** current optimizations, performance features **Persona:** Developer

Difficulty: Advanced

[PERF-02] Q: What performance improvements are planned but not yet implemented?

Intent: Informational — performance & scalability

Answer: Documented potential improvements include Redis caching for frequently accessed data, pagination on list endpoints, further database query optimization, file compression for uploads, API rate limiting, and more thorough transaction management for critical operations.

Alternative phrasings: just wondering, what performance improvements are planned but not yet implemented?; What performance improvements are planned but not yet implemented?; hey carebot, what performance improvements are planned but not yet implemented; planned performance work?

Possible follow-ups: Which list endpoints most need pagination first? | Would Redis caching apply to dashboard aggregates?

Related topics: PERF-01, GAP-01 **Keywords:** planned performance work, future scalability improvements **Persona:**

Developer **Difficulty:** Advanced

5.x Data & Migration

[DATA-01] Q: How does CareBridge handle database migrations?

Intent: Informational — data & migration

Answer: In development, Hibernate's auto-DDL feature manages schema changes automatically. For production, the knowledge base recommends using a versioned migration tool such as Flyway or Liquibase, alongside a regular MySQL backup strategy.

Alternative phrasings: so how does CareBridge handle database migrations?; How does CaerBridge handle database migrations?; hey carebot, how does carebridge handle database migrations; database migrations?

Possible follow-ups: Is Flyway or Liquibase actually wired in today, or just recommended? | How frequently are production backups taken?

Related topics: DATA-02, TECH-04 **Keywords:** database migrations, flyway liquibase, schema changes **Persona:** Developer

Difficulty: Advanced

[DATA-02] Q: Why does CareBridge keep rejected profiles and deleted items instead of just deleting them?

Intent: Informational — data & migration

Answer: Rejected donor and orphanage profiles are archived into RejectedDonorProfile and RejectedOrpProfile, and deleted need items go into DeletedItems, specifically to provide an audit trail and prevent silent data loss, rather than discarding records outright when they leave the active dataset.

Alternative phrasings: so why does CareBridge keep rejected profiles and deleted items instead of just deleting them?; Why does CareBridge keep rejected profiles and deleted items instead of just deleting them?; hey carebot, why does carebridge keep rejected profiles and deleted items instead of just deleting them; why archive data?

Possible follow-ups: Is there a retention limit on these archive tables? | Can archived data be permanently purged on request?

Related topics: TECH-05, GAP-01 **Keywords:** why archive data, audit trail purpose **Persona:** Developer **Difficulty:** Intermediate

5.x Frontend Architecture

[FE-01] Q: How many pages and components does the CareBridge frontend have?

Intent: Informational — frontend architecture

Answer: The donor/orphanage frontend has 41+ page components and 33+ reusable components, organized alongside 13 dedicated API integration files, custom route guards, and utility modules like `authUtils.js` and `categoryIcons.js`.

Alternative phrasings: quick q - how many pages and components does the CareBridge frontend have?; How many pages and components does the CareBridge frontend have?; hey carebot, how many pages and components does the carebridge frontend have; frontend page count?

Possible follow-ups: Which pages are shared between donor and orphanage roles, if any? | How is routing protected per role?

Related topics: FE-02, TECH-01 **Keywords:** frontend page count, component count, frontend structure **Persona:** Developer **Difficulty:** Advanced

[FE-02] Q: How does the frontend manage authentication state and protected routes?

Intent: Informational — frontend architecture

Answer: `authUtils.js` handles JWT token storage and retrieval from `localStorage`, detects the user's role from the decoded token (via `jwt-decode`), and provides login/logout helpers plus a protected-route wrapper used by `AppRoutes.jsx` to guard donor, orphanage, and admin routes.

Alternative phrasings: so how does the frontend manage authentication state and protected routes?; How does the frontend manage authentication state and protected routes?; hey carebot, how does the frontend manage authentication state and protected routes; frontend auth state?

Possible follow-ups: What happens if a decoded token has an unexpected role value? | Is the token also stored in a cookie, or only `localStorage`?

Related topics: FE-01, AUTH-01 **Keywords:** frontend auth state, protected routes, `jwt-decode` **Persona:** Developer **Difficulty:** Advanced

[FE-03] Q: How are category icons displayed for need items on the frontend?

Intent: Informational — frontend architecture

Answer: `categoryIcons.js` maps each `CategoryType` enum value (FOOD, EDUCATION, HEALTH, CLOTHING, TOYS, SPORTS, TECHNOLOGY, OTHER) to a corresponding React icon, which is used wherever need items are rendered, such as `NeedCard.jsx` and `NeedsGrid.jsx`.

Alternative phrasings: quick q - how are category icons displayed for need items on the frontend?; How are category icons displayed for need items on the frontend?; hey carebot, how are category icons displayed for need items on the frontend; category icons?

Possible follow-ups: What icon is shown for the OTHER category? | Are these icons from React Icons or Lucide React?

Related topics: NEED-02, FE-01 **Keywords:** category icons, need item icons, categoryIcons.js **Persona:** Developer
Difficulty: Intermediate

[FE-04] Q: What does the CareBot chatbot component structure look like on the frontend?

Intent: Informational — frontend architecture

Answer: The chatbot lives in a dedicated `src/components/chatbot/` directory with eight files: `CareBot.jsx` (main component), `ChatWindow.jsx` (display area), `ChatMessage.jsx` (individual message), `ChatInput.jsx` (input field), `SuggestedQuestions.jsx` (predefined prompts), `TypingIndicator.jsx` (typing animation), `carebotApi.js` (backend API calls), and `chatbot.css` (styling).

Alternative phrasings: quick q - what does the CareBot chatbot component structure look like on the frontend?; What does the CareBot chatbot component structure look like on the frontend?; hey carebot, what does the carebot chatbot component structure look like on the frontend; chatbot component structure?

Possible follow-ups: Where does `carebotApi.js` send its requests — a separate RAG service or the main backend? | Are `SuggestedQuestions.jsx` prompts drawn from this documentation?

Related topics: FE-01, RAG-DOC **Keywords:** chatbot component structure, carebot files **Persona:** Developer **Difficulty:** Intermediate

5.x Admin Portal

[ADMPORT-01] Q: Is the admin portal the same application as the main CareBridge website?

Intent: Informational — admin portal

Answer: No. `CarebridgeAdminPortal` is a fully separate React 19.2.6 + Vite 8.0.12 application with its own API layer, routing, and deployment, though it communicates with the same Spring Boot backend as the donor/orphanage frontend.

Alternative phrasings: quick q - is the admin portal the same application as the main CareBridge website?; Is the admin portal the same application as the main CareBridge website?; hey carebot, is the admin portal the same application as the main carebridge website; separate admin app?

Possible follow-ups: Can the admin portal be hosted on a different domain? | Do admins log in with the same credentials as donors or orphanages?

Related topics: ADMPORT-02, TECH-01 **Keywords:** separate admin app, admin portal vs main site **Persona:** Platform Admin **Difficulty:** Intermediate

[ADMPORT-02] Q: What pages exist in the admin portal?

Intent: Informational — admin portal

Answer: Key pages include `AdminLoginPage.jsx`, `PendingDonorsPage.jsx`, `PendingOrphanagesPage.jsx`, and organized subdirectories for auth, dashboard, donations, donors, orphanages, settings, and visits, plus a `GlobalSlotsPage.jsx` for viewing every slot across the whole platform.

Alternative phrasings: hey, what pages exist in the admin portal?; What pages exist in the admin portal?; hey carebot, what pages exist in the admin portal; admin portal pages?

Possible follow-ups: Does `GlobalSlotsPage` let an admin cancel a slot directly? | Is there a settings page for CORS or SMTP configuration?

Related topics: ADMPORT-01, SLOT-01 **Keywords:** admin portal pages, admin navigation **Persona:** Platform Admin **Difficulty:** Basic

[ADMPORT-03] Q: What shared components does the admin portal use for reviewing profiles?*Intent:* Informational — admin portal

Answer: AdminSidebar.jsx provides main navigation, LoadingSpinner.jsx shows loading state, and RejectModal.jsx handles the rejection-confirmation flow (capturing the rejection reason before calling the reject endpoint).

Alternative phrasings: quick q - what shared components does the admin portal use for reviewing profiles?; What shared components does teh admin portal use for reviewing profiles?; hey carebot, what shared components does the admin portal use for reviewing profiles; admin components?

Possible follow-ups: Does RejectModal enforce a minimum reason length? | Is there a bulk-reject option in the current admin UI?

Related topics: ADMPORT-02, VERIF-02 **Keywords:** admin components, rejectmodal, adminsidebar **Persona:** Platform Admin **Difficulty:** Basic

5.x Integrations

[INT-01] Q: What third-party services does CareBridge integrate with today?*Intent:* Informational — integrations

Answer: Today's integrations are: email delivery (Gmail SMTP in development, Brevo SMTP API in production) and image storage (Cloudinary, for profile pictures, documents, and donation/delivery photos). Authentication is self-contained via JWT with no third-party identity provider, and there is no payment integration yet.

Alternative phrasings: so what third-party services does CareBridge integrate with today?; What thrid-party services does CareBridge integrate with today?; hey carebot, what third-party services does carebridge integrate with today; third party integrations?

Possible follow-ups: Is there any plan to add an identity provider like Google Sign-In? | Which integration is most likely to be added next?

Related topics: INT-02, DON-08 **Keywords:** third party integrations, external services used **Persona:** Developer **Difficulty:** Advanced

[INT-02] Q: What future integrations are planned for messaging?*Intent:* Informational — integrations

Answer: Beyond today's email-based notifications, the roadmap lists SMS delivery via Twilio and WhatsApp integration as near-term additions for reaching donors and orphanages on more immediate channels.

Alternative phrasings: so what future integrations are planned for messaging?; What fuutre integrations are planned for messaging?; hey carebot, what future integrations are planned for messaging; future messaging integrations?

Possible follow-ups: Would SMS replace email OTPs or supplement them? | Is WhatsApp integration meant for notifications, two-way messaging, or both?

Related topics: INT-01, NOTIF-01 **Keywords:** future messaging integrations, twilio whatsapp **Persona:** Developer **Difficulty:** Intermediate

5.x Chatbot Corpus Summary

This chapter contains 97 distinct topic entries and 485 total question surface-forms (canonical plus generated variants) across 22 categories, spanning personas from first-time visitors to developers. Chapter 6 supplements this corpus with a shorter-form FAQ repository, and Chapter 8 explains how to chunk and index both chapters for a Qdrant-backed RAG pipeline.

Chapter 6 — Complete FAQ Repository

This chapter presents CareBridge's facts in short-form question-and-answer pairs — ideal for a help-center page, a searchable FAQ widget, or as short high-confidence chunks in a vector database. It restates the Chapter 5 topics in condensed form, organized by category, and adds further FAQs not covered as full chatbot entries.

6.1 Registration & Onboarding

1. How do I register on CareBridge?

Registration starts at POST /auth/register, which generates a One-Time Password (OTP) and emails it to the address you supplied. You then call POST /auth/verify with that OTP to confirm your identity and create your user account. If the OTP does not arrive in time, POST /auth/resend issues a new one (a five-minute expiry window applies to each OTP).

2. Is registering an account the same as creating my donor or orphanage profile?

No. Registration (POST /auth/register + POST /auth/verify) only creates a basic User record with a name, email, hashed password, and role (DONOR, ORPHANAGE, or ADMIN). A separate step — POST /donor/create-profile or POST /orphanage/create-profile — captures the detailed information (PAN, address, documents, etc.) needed before you can use the platform.

3. Why did my registration fail?

The most common causes are: the email is already registered (emails must be unique per account), the email format is invalid, the password does not meet strength requirements, a temporary server/connection issue occurred, or a form validation rule was not met. Checking the on-screen validation messages usually identifies the exact field that needs correcting.

4. How long is the OTP valid for during registration?

Every OTP issued by CareBridge, including the one sent during registration, expires five minutes after it is generated. If it lapses, request a new code through the resend endpoint rather than retrying the expired one.

5. Does orphanage registration require OTP verification too?

Yes. After submitting POST /orphanage/create-profile, the orphanage must call POST /orphanage/verifyOrpOTP to confirm the OTP emailed to the orphanage's registered address. POST /orphanage/resendOrpOTP is available if the original OTP is not received or expires.

6.2 Authentication & Security

6. How do I log in to CareBridge?

Call POST /auth/login with your registered email and password. On success the API returns a JWT access token valid for 24 hours. That token must be sent as an 'Authorization: Bearer {token}' header on every subsequent protected request; CareBridge uses stateless JWT authentication, so no server-side session is stored.

7. Why do I keep getting logged out / getting a 401 error?

A 401 Unauthorized response on a protected endpoint almost always means your JWT has expired. Tokens are valid for 24 hours from login; once that window passes you must log in again via POST /auth/login to obtain a fresh token.

8. I forgot my password — how do I reset it?

Go to the forgot-password flow: POST /auth/forgot-password sends an OTP to your registered email, POST /auth/verify-forgot-otp checks that OTP, and POST /auth/reset-password lets you set a new password. Afterward, log in normally with the new password.

9. How are passwords and roles secured on the backend?

Passwords are hashed with BCrypt and never stored in plain text. Role-based access control restricts endpoints by ROLE_DONOR, ROLE_ORPHANAGE, and ROLE_ADMIN in SecurityConfig, and a JwtAuthFilter intercepts every request, validates the token's signature and expiry, and populates the SecurityContext. Session policy is STATELESS — every request must carry a valid JWT rather than relying on a server-side session.

10. Which API endpoints can be accessed without logging in?

SecurityConfig permits unauthenticated access to /auth/** (registration, login, verification), /api/landing/** (public landing page data), and the API documentation routes /swagger-ui/** and /v3/api-docs/**. Every other route requires a valid JWT and the correct role.

6.3 Donor Profile

11. What information do I need to create my donor profile?

POST /donor/create-profile (multipart/form-data) requires your name, date of birth, designation, phone number, PAN number, and a PAN photo upload. Optional fields include a profile picture and address details (house number, mandal, village, district, state, country). The system then generates a unique CareBridgeID for your profile and sets its status to PENDING.

12. What format does the PAN number need to follow?

The PAN number is validated against the standard Indian pattern of five letters, four digits, and one letter (e.g., AAAAA9999A). It must also be unique — the system rejects a PAN number already tied to another profile.

13. Can I update my donor profile after it's been submitted?

Yes. PUT /donor/update-profile (multipart/form-data) lets you update fields such as designation, address, phone, pictures, and PAN. Updates are possible both before and after admin approval; simple field changes do not require re-approval, though sensitive fields like PAN or date of birth may warrant another review.

14. What is a CareBridgeID and where do I get one?

A CareBridgeID is a unique identifier auto-generated by the backend for every donor and orphanage profile. It is distinct from your internal user ID and is the reference used across donations, needs, and slot bookings. You don't choose it — the system assigns it automatically when your profile is created.

6.4 Orphanage Profile

15. What documents does an orphanage need to submit to register?

POST /orphanage/create-profile requires the orphanage name, admin name, number of children, email, phone, district, state, country, a DARPAN ID (social-sector registration), a PAN number with PAN photo, and a JJ Act Certificate photo (proof of Juvenile Justice Act compliance). Village, mandal, website link, social media links, and profile pictures are optional.

16. What is DARPAN ID and why does CareBridge need it?

DARPAN ID is the registration number issued under India's social-sector scheme, and CareBridge requires it as one of the mandatory documents proving an orphanage is a legitimate, registered organization before it can be verified and allowed to post needs or accept donations.

6.5 Verification & Admin Approval

17. Why is my profile still showing as pending?

PENDING means your profile has been created and submitted but has not yet been reviewed by a platform admin. Approval is a manual step: an admin views your documents via GET /admin/donor/pending (or the orphanage equivalent) and then approves or rejects the profile. If it has been pending an unusually long time, that typically means the admin queue simply hasn't reached it yet.

18. How does an admin approve or reject a donor or orphanage profile?

Admins call PATCH /admin/donor/approve/{id} or PATCH /admin/orphanage/approve/{id} to set status to VERIFIED and trigger an approval email. To reject, PATCH /admin/donor/reject/{id} or the orphanage equivalent accepts an EmailReason payload; this moves the profile into the RejectedDonorProfile or RejectedOrpProfile archive table, deletes the original profile record, and emails the reason to the applicant.

19. Why was my profile rejected?

Common rejection reasons include an invalid PAN format or a blurry PAN photo, a PAN mismatch against your stated name or date of birth, a PAN number already registered by someone else, invalid or unclear supporting documents, or generally suspicious or incomplete information. Rejections are permanent for that submission — you must register again with corrected details rather than editing the rejected record.

20. What's the difference between email verification and profile verification?

Email verification simply confirms you own the email address you registered with, and happens during the OTP step of registration. Profile verification is a separate, later step: an admin reviews your submitted documents and sets your status to PENDING, VERIFIED, or REJECTED. You can be email-verified while your profile is still PENDING.

21. Can I donate before my profile is verified?

No. The required sequence is: register and verify your email via OTP, create your full profile with PAN and personal details, and then wait for an admin to approve it (you'll receive a VERIFIED-status confirmation email). Only after that can you donate or book visits.

6.6 Need Items & Categories

22. How do I post a need for my orphanage?

Call POST /orphanage/needItem/create-item with a name (e.g. 'School Uniforms'), a category, quantity, price per quantity, priority, and optionally a description and product links. This creates a NeedItem with a unique needItemId, isActive set to true, and fulfilledQuantity starting at zero.

23. What categories of needs can orphanages post?

The `CategoryType` enum covers `FOOD`, `EDUCATION` (books, school supplies, uniforms), `HEALTH` (medicines, medical equipment), `CLOTHING`, `TOYS` (recreational items), `SPORTS` (sports equipment), `TECHNOLOGY` (computers, tablets, internet access), and `OTHER`.

24. What do the priority levels LOW, MEDIUM, HIGH, and URGENT mean for a need?

Priority signals how critical a need is: `LOW` means 'nice to have', `MEDIUM` means 'important', `HIGH` means 'significant need', and `URGENT` means 'critical/immediate need'. These labels help donors decide which needs to prioritize and support impact messaging on the platform.

25. Can more than one donor contribute to the same need?

Yes. CareBridge tracks total quantity needed, `fulfilledQuantity` (already delivered), and `reservedQuantity` (pledged by donors who haven't delivered yet). Remaining quantity is calculated as total minus fulfilled minus reserved, so multiple donors can each cover a portion of the same need until it is fully met.

26. How do I edit or remove a need item I posted?

`PUT /orphanage/needItem/updateItem/{id}` lets you modify name, quantity, price, priority, description, and product links (the `needItemId` and owning orphanage cannot be changed). `DELETE /orphanage/needItem/deleteItem/{id}` removes it — typically a soft delete that sets `isActive` to false, with a copy retained in the `DeletedItems` table for audit history.

27. Where can I see all the needs my orphanage has posted?

`GET /orphanage/needItem/activeNeeds` lists all needs your orphanage has posted, along with fulfillment progress for each item so you can track how close every need is to being fully met.

28. How is a unique needItemId generated?

A dedicated utility class, `NeedItemIdGenerator`, is responsible for producing each unique `needItemId` when a new `NeedItem` is created by `NeedItemService`.

6.7 Donations

29. How do I make a donation on CareBridge?

Call `POST /donor/donate` with the `needItemId` you're supporting, a quantity, a `donationType` (`IN_PERSON` or `ONLINE_ORDER`), and type-specific details (`expectedVisitDateTime` for in-person, or `expectedDeliveryDate`, `platformName`, `trackingId`, and an order proof image for online orders). The donation is created with status `PENDING` and returns a `donationRequestId`.

30. What's the difference between an IN_PERSON donation and an ONLINE_ORDER donation?

`IN_PERSON` donations mean you deliver the item yourself during a scheduled visit; you must book a slot through the visit-booking system and provide an `expectedVisitDateTime`. `ONLINE_ORDER` donations mean you purchase the item online (e.g., from Amazon) and have it shipped; you supply an `expectedDeliveryDate`, the `platformName`, a `trackingId`, and an `orderProofImage` instead of booking a slot.

31. Can I cancel a donation I already made?

Only while it is still `PENDING` — `DELETE /donor/delete-pending-donation/{id}` removes it and frees up the reserved quantity for other donors. Once a donation's status is `DELIVERED` it cannot be cancelled, because the item has already been received and confirmed by the orphanage.

32. Can I edit a donation after I've submitted it?

Yes, via `PATCH /donor/update/{donationId}`, but only while the donation is still `PENDING`. You can change quantity, donation type, visit date, delivery date, platform, tracking ID, message, and the order proof image. You cannot switch which need item the donation is tied to — that requires creating a new donation. Once `DELIVERED`, nothing can be modified.

33. Where can I see the status of my past donations?

Four donor-side endpoints cover this: `GET /donor/pending-donations` for donations awaiting delivery, `GET /donor/completed-donations` and `GET /donor/delivered-donations` for delivered items, and `GET /donor/cancelled-donations` for cancelled ones.

34. How does an orphanage confirm that a donation was delivered?

`POST /orphanage/delivered/{id}` accepts an `ItemDeliveredDTO` (multipart) with an optional delivery note and a delivered-photo upload. This sets the donation's status to `DELIVERED`, records the delivery date, note, and photo URL, and triggers an email notification to the donor.

35. What happens if a delivery to us fails or the items never show up?

If a delivery fails, the orphanage can decline to confirm it; the donation then returns to `PENDING` status rather than being marked `DELIVERED`, so it still appears in the pending queue for follow-up or eventual cancellation by the donor.

36. Does CareBridge process payments for donations?

No. CareBridge is currently not a payment platform. Donations take the form of in-person item delivery or online-ordered item delivery; any actual payment (for example, paying Amazon for the item) is arranged separately by the donor outside the platform.

37. Are my donations anonymous to the orphanage?

No. CareBridge is built around transparency and accountability, so the receiving orphanage can see your name and profile information, and you can see the orphanage's details in turn. Donors also receive impact notifications tied to their donations.

38. What are all the possible donation status values?

The documented `DonationStatus` values are `PENDING` (created, awaiting delivery), `DELIVERED` (received and confirmed by the orphanage), and `CANCELLED` (cancelled by the donor or system). A `COMPLETED` status is noted as a possible alternate status for fully fulfilled donations, though its exact usage alongside `DELIVERED` is not fully detailed in the knowledge base.

6.8 Visit Slots & Booking

39. How do I create visit slots for donors as an orphanage?

`POST /orphanage/slot/create` takes a `slotDate`, `startTime`, `endTime`, and `capacity` (maximum visitors), and returns a `SlotResponse` with the new `slotId` and status. You can create multiple slots across different dates and times; each is a one-off window rather than a recurring schedule.

40. How do I find available visit slots for a specific orphanage?

Call `GET /donor/slots/available-slots/{orphanageCareBridgeId}` to list every `AVAILABLE` slot for that orphanage, including date, time, capacity, and remaining open spots.

41. What do the slot statuses AVAILABLE, EXPIRED, CANCELLED, and FULL mean?

AVAILABLE means the slot can still be booked. EXPIRED means its scheduled time has already passed — this is updated automatically by SlotSchedulerService, which runs scheduled jobs to expire past slots and clean up stale bookings. CANCELLED means the orphanage withdrew the slot. FULL means every capacity spot has been reserved.

42. How do I book a visit to an orphanage?

POST /donor/slots/book takes the slotId, the number of visitors, and an optional message. The system validates that the number of visitors doesn't exceed remaining capacity, the slot is still AVAILABLE, and your donor profile is VERIFIED. A successful booking is created with status PENDING and returns a bookingId.

43. How do I know if my visit booking is confirmed?

After you book, the orphanage reviews it and calls PATCH /orphanage/slot/confirm/{bookingId} to set status to CONFIRMED, or PATCH /orphanage/slot/reject/{bookingId} with a reason to set it to REJECTED. You'll be notified of the decision either way.

44. How does an orphanage manage incoming visit booking requests?

GET /orphanage/slot/pending lists every PENDING visit booking, showing donor details and visitor counts, so the orphanage can decide whether to confirm or reject each one. Notifications are sent to the donor once a decision is made.

45. Can I reuse the same booked slot for more than one visit?

No — one slot booking corresponds to one visit opportunity. Multiple donations can occur during that single visit, but if you want to visit again you need to book a separate slot for a new date and time.

46. What's the difference between a slot and a donation?

A slot is a time window an orphanage makes available for visits; a donation is the actual transfer of items or value that a donor initiates. They work together for in-person giving — you must book a slot before an IN_PERSON donation visit — but ONLINE_ORDER donations don't require a slot at all.

47. What automated jobs keep slot data accurate over time?

SlotSchedulerService runs scheduled (cron-style) tasks that mark slots EXPIRED once their scheduled time has passed, clean up cancelled or finished bookings, and update availability counts, so donors never see a stale slot as bookable.

6.9 Dashboards & Analytics

48. What does my donor dashboard show me?

The donor dashboard combines a profile summary (verification and subscription status, member-since date), donation statistics (counts, total value, breakdown by category and status), an impact summary (orphanages helped, estimated lives impacted, fulfillment contribution), your most recent donations, upcoming confirmed bookings, and a list of urgent needs recommended to you.

49. How is my donor impact score or 'lives impacted' calculated?

Impact is derived from delivered (not pending) donations: total donations delivered, their total monetary value, the number of distinct orphanages helped, and the categories of needs fulfilled feed into an estimated lives-impacted figure — for example, fifty delivered uniforms translates to roughly fifty children impacted.

50. What does the orphanage dashboard display?

It provides an overview (total needs posted, total donations received, total visitors, overall fulfillment percentage), a donation summary (counts, value, breakdown by category and by IN_PERSON vs ONLINE_ORDER, recent timeline, pending count), a needs summary (active needs, quantities needed vs. fulfilled, average fulfillment by category), a visit summary (confirmed, upcoming, and past visits, average visitors per visit), and an activity feed of recent donations, new needs, and upcoming confirmations.

51. Where do I see my upcoming scheduled visits?

The donor dashboard's 'Upcoming Bookings' section lists your next scheduled visits with slot details (date, time, orphanage), confirmation status, and reminder notifications.

52. What can a platform admin see on their dashboard?

The admin dashboard shows overall platform statistics, the current count of pending approvals, recent activity across the system, and system health metrics, alongside separate views for pending, approved, and rejected donor and orphanage profiles.

53. How does CareBridge recommend needs to me?

The donor dashboard surfaces 'Urgent Needs' — high-priority needs from verified orphanages, prioritizing orphanages in greatest need and matching against criteria relevant to donor interests.

6.10 Notifications & OTP

54. What is an OTP used for on CareBridge?

OTPs (One-Time Passwords) are six-digit codes emailed for registration verification, password resets, and orphanage verification. Each OTP is stored with a five-minute expiry and a used flag, and becomes invalid once used or expired.

55. I never received my OTP email — what should I do?

First check your spam or junk folder, since automated verification emails are sometimes filtered there. If it's still missing, use the resend endpoint to generate a new OTP (allowed roughly three to five times before a cooldown applies), and confirm the email address you entered was correct.

56. What kinds of emails will CareBridge send me?

Email notifications are triggered for: registration OTPs, password-reset OTPs, orphanage-registration OTPs, profile approval, profile rejection (with reason), donation creation confirmation, donation delivery confirmation, and visit booking confirmation or rejection.

57. What email infrastructure does CareBridge use to send notifications?

Development environments send mail through Gmail SMTP, while production uses the Brevo SMTP API, both configured through Spring Mail properties in application-dev.properties and application-prod.properties respectively. EmailMessages.java acts as a template library for the formatted content of each notification type.

6.11 Messaging

58. Can I message an orphanage directly through CareBridge?

Yes, messaging components exist on both sides of the platform — `DonorMessages.jsx` and `OrphanageMessagePage.jsx` on the frontend, backed by a shared `Messages.jsx` component — allowing donors and orphanages to communicate within the platform.

6.12 Landing Page & Browse

59. Can I see CareBridge's impact statistics without creating an account?

Yes. `GET /api/landing/stats` is a public endpoint (no authentication needed) that returns global statistics such as total verified orphanages, total active donors, total donations received, total items distributed, and success-story counts, so visitors can see the platform's impact before registering.

60. Can I browse orphanages and needs before registering?

Yes. `GET /api/landing/orphanages` lists verified orphanages (name, photo, location, number of children, active needs, total donations, recent activity), and `GET /api/landing/needs` lists active needs — both are public endpoints intended for discovery and marketing before you create an account.

6.13 Admin Operations

61. What can a platform admin actually do on CareBridge?

Admins can approve or reject donor and orphanage profiles, view overall platform statistics, monitor donations and visit bookings, manage system settings, view user data for verification purposes, and send notifications. Admins do not create donations as a donor, post needs as an orphanage, or book visits themselves.

62. What criteria do admins use when reviewing a profile?

For donors, admins check for a valid PAN number format, a clear PAN photo, valid contact information, and no duplicate records. For orphanages, they check valid DARPAN registration, a valid JJ Act Certificate, a valid PAN number, legitimate contact information, and a verifiable registered address.

6.14 Technical Architecture

63. What is the overall technology stack behind CareBridge?

The backend is Spring Boot 4.0.6 on Java 21 with Maven, using MySQL in production and H2 for testing, following a layered Controller-Service-Repository-Entity architecture with Hibernate as the ORM and Swagger/OpenAPI documentation. The donor/orphanage frontend is React 19.2.5 with Vite 8.0.10 and Bootstrap 5.3.8; the admin portal is a separate React 19.2.6 + Vite 8.0.12 application.

64. How is the backend organized into modules?

The backend is split into twelve functional modules: authentication, donor profile management, orphanage profile management, donation management, visit-booking management, needs management, dashboard management (donor and orphanage), notification management, admin management, slot-details fetch management, landing-page data management, and common utilities (Cloudinary config, image upload, the shared `ApiResponse` wrapper).

65. How does CareBridge handle image and file uploads?

ImageUploadService routes all uploads (profile pictures, PAN photos, certificates, donation and delivery photos) to Cloudinary via CloudinaryConfig, supporting JPG, PNG, and GIF up to 20MB per file and 100MB per request. Only the resulting Cloudinary URL is stored in the database — images themselves live in Cloudinary's CDN and are not deleted even if the related profile is later removed.

66. What are the core database entities and how do they relate?

User relates 1:1 to DonorProfile and 1:1 to OrphanageProfile. OrphanageProfile relates 1:many to NeedItem and 1:many to Slot. DonorProfile relates 1:many to DonationRequest and 1:many to VisitBooking. NeedItem relates 1:many to DonationRequest, and Slot relates 1:many to VisitBooking. Archive tables (RejectedDonorProfile, RejectedOrpProfile, DeletedItems) preserve history when records are rejected or removed.

67. How is data archived when a profile is rejected or a need item is deleted?

Rejected donor and orphanage profiles are copied into RejectedDonorProfile and RejectedOrpProfile respectively (with the rejection reason attached) before the original record is deleted. Deleted need items are copied into the DeletedItems table with deletedBy, deletionReason, and deletedAt fields, preserving an audit trail rather than silently discarding data.

68. What security practices are already implemented versus still planned?

Implemented: BCrypt password hashing, stateless JWT authentication, role-based access control, CORS configuration, @Valid input validation, exception handling with error masking, multipart upload validation, OTP expiry/single-use enforcement, and ORM-based SQL-injection prevention. Still to implement: enforced HTTPS/SSL, API rate limiting, request logging and monitoring, two-factor authentication, audit logging for sensitive operations, encryption at rest, GDPR-style data deletion, and additional security headers such as X-Frame-Options and CSP.

69. How do I run CareBridge locally for development?

Backend: install a Java 21 SDK and Maven, run a local MySQL server, then run 'mvn spring-boot:run'. Frontend: install Node.js 18+, then run 'npm run dev' to start the Vite dev server (typically on port 5173). The admin portal follows the same Node.js/npm pattern as a separate app.

70. How is CareBridge deployed to production?

The backend can be deployed to any cloud host supporting a Java 21 runtime (AWS, Azure, Heroku, DigitalOcean) with a MySQL database and environment variables for database credentials, the JWT secret, SMTP credentials, Cloudinary keys, and CORS origins, built via 'mvn clean install' and run as a jar. The frontend and admin portal are each built with 'npm run build' and deployed independently to platforms like Firebase Hosting, Vercel, or Netlify.

71. What does the standard CareBridge API response look like?

ApiResponse.java defines a consistent wrapper used across endpoints, with a boolean 'success' field, a 'message' string, and a 'data' object carrying the actual payload, ensuring every response follows the same shape regardless of which module handled the request.

72. Where can I find the full, always-current CareBridge API specification?

The backend auto-generates OpenAPI documentation via springdoc-openapi-starter-webmvc-ui, browsable at /swagger-ui/** and machine-readable at /v3/api-docs/**, both public routes that don't require authentication.

73. What AI stack powers the CareBot chatbot's answers?

CareBot uses Retrieval-Augmented Generation: an OpenRouter-hosted LLM generates the reply text, grounded by passages retrieved from a Qdrant Cloud vector database that stores embeddings of the platform's knowledge base (see Chapter 8 for chunking and retrieval guidance).

6.15 Troubleshooting

74. I'm getting a CORS error in the browser — what does that mean?

A CORS error ('Access to XMLHttpRequest blocked by CORS policy') means the frontend's origin isn't in the backend's allowed-origins list. The fix is on the backend configuration side: confirm the frontend URL is whitelisted in the CORS settings.

75. Why do I suddenly get 'Unauthorized' errors while using the app?

This is the standard symptom of an expired JWT (tokens last 24 hours from login). Logging in again via POST /auth/login issues a fresh token and resolves it.

76. My file upload keeps failing with a 413 error — why?

A 413 'Payload Too Large' response means the file exceeds the 20MB per-file limit, or the overall multipart request exceeds the 100MB total limit configured on the backend. Compress or resize the file and try again.

77. My profile has been PENDING for a long time — is something broken?

Not necessarily. PENDING simply means an admin hasn't reviewed your profile yet; there's no automatic timeout. If it's been an unusually long wait, the practical next step is to check in with the platform admin directly.

78. I can't create a donation — it says the orphanage wasn't found or quantity is insufficient. What's wrong?

Double-check that the needItemId you're referencing is correct, that the need's remaining quantity is not already zero, and that the target orphanage's profile is VERIFIED — donations against an unverified or not-found orphanage/need will be rejected by the backend.

79. Why does booking an available slot keep failing for me?

Slot booking fails if your donor profile isn't yet VERIFIED, if the requested capacity has already been used up by other bookings, or if the slot's date/time has already passed. Confirm your verification status first, since that's the most common blocker.

6.16 Policies & Trust

80. How does CareBridge make sure orphanages are legitimate?

Every orphanage must submit a DARPAN registration ID, a PAN number with photo, and a JJ Act Certificate proving Juvenile Justice Act compliance. A platform admin manually reviews these documents before granting VERIFIED status, and only verified orphanages can post needs or receive donations.

81. How does CareBridge verify that donors are real people?

Donors must provide a valid PAN number in the standard Indian format along with a PAN photo, plus personal details like date of birth, designation, and a unique phone number. An admin reviews this before granting VERIFIED status, which is required before the donor can donate or book visits.

6.17 Testing & Quality Assurance

82. What testing tools and commands does the CareBridge backend use?

Backend tests use Mockito for unit-level mocking and an H2 in-memory database for integration and repository tests, covering repository, service, and controller layers. Run the suite with 'mvn test', or 'mvn verify' for a full verification build.

83. How is the frontend tested?

Component-level testing uses React Testing Library, while end-to-end flows are tested with Playwright or Cypress. Manual cross-browser verification is also performed across Chrome, Firefox, and Safari. ESLint enforces code quality, and SonarQube is listed as an optional static-analysis addition.

84. What tool checks code quality on the frontend?

ESLint is used for frontend code quality; SonarQube is mentioned as an optional tool for deeper code analysis, alongside JUnit for backend testing.

6.18 Performance & Scalability

85. What performance optimizations does CareBridge already have in place?

Stateless JWT authentication avoids server-side session storage, frequently accessed fields (email, phone, CareBridgeID) are indexed, frontend components are lazy-loaded, images are optimized and CDN-delivered via Cloudinary, and Spring Data JPA manages database connection pooling.

86. What performance improvements are planned but not yet implemented?

Documented potential improvements include Redis caching for frequently accessed data, pagination on list endpoints, further database query optimization, file compression for uploads, API rate limiting, and more thorough transaction management for critical operations.

6.19 Data & Migration

87. How does CareBridge handle database migrations?

In development, Hibernate's auto-DDL feature manages schema changes automatically. For production, the knowledge base recommends using a versioned migration tool such as Flyway or Liquibase, alongside a regular MySQL backup strategy.

88. Why does CareBridge keep rejected profiles and deleted items instead of just deleting them?

Rejected donor and orphanage profiles are archived into RejectedDonorProfile and RejectedOrpProfile, and deleted need items go into DeletedItems, specifically to provide an audit trail and prevent silent data loss, rather than discarding records outright when they leave the active dataset.

6.20 Frontend Architecture

89. How many pages and components does the CareBridge frontend have?

The donor/orphanage frontend has 41+ page components and 33+ reusable components, organized alongside 13 dedicated API integration files, custom route guards, and utility modules like authUtils.js and

categoryIcons.js.

90. How does the frontend manage authentication state and protected routes?

authUtils.js handles JWT token storage and retrieval from localStorage, detects the user's role from the decoded token (via jwt-decode), and provides login/logout helpers plus a protected-route wrapper used by AppRoutes.jsx to guard donor, orphanage, and admin routes.

91. How are category icons displayed for need items on the frontend?

categoryIcons.js maps each CategoryType enum value (FOOD, EDUCATION, HEALTH, CLOTHING, TOYS, SPORTS, TECHNOLOGY, OTHER) to a corresponding React icon, which is used wherever need items are rendered, such as NeedCard.jsx and NeedsGrid.jsx.

92. What does the CareBot chatbot component structure look like on the frontend?

The chatbot lives in a dedicated src/components/chatbot/ directory with eight files: CareBot.jsx (main component), ChatWindow.jsx (display area), ChatMessage.jsx (individual message), ChatInput.jsx (input field), SuggestedQuestions.jsx (predefined prompts), TypingIndicator.jsx (typing animation), carebotApi.js (backend API calls), and chatbot.css (styling).

6.21 Admin Portal

93. Is the admin portal the same application as the main CareBridge website?

No. CarebridgeAdminPortal is a fully separate React 19.2.6 + Vite 8.0.12 application with its own API layer, routing, and deployment, though it communicates with the same Spring Boot backend as the donor/orphanage frontend.

94. What pages exist in the admin portal?

Key pages include AdminLoginPage.jsx, PendingDonorsPage.jsx, PendingOrphanagesPage.jsx, and organized subdirectories for auth, dashboard, donations, donors, orphanages, settings, and visits, plus a GlobalSlotsPage.jsx for viewing every slot across the whole platform.

95. What shared components does the admin portal use for reviewing profiles?

AdminSidebar.jsx provides main navigation, LoadingSpinner.jsx shows loading state, and RejectModal.jsx handles the rejection-confirmation flow (capturing the rejection reason before calling the reject endpoint).

6.22 Integrations

96. What third-party services does CareBridge integrate with today?

Today's integrations are: email delivery (Gmail SMTP in development, Brevo SMTP API in production) and image storage (Cloudinary, for profile pictures, documents, and donation/delivery photos). Authentication is self-contained via JWT with no third-party identity provider, and there is no payment integration yet.

97. What future integrations are planned for messaging?

Beyond today's email-based notifications, the roadmap lists SMS delivery via Twilio and WhatsApp integration as near-term additions for reaching donors and orphanages on more immediate channels.

6.23 Additional Platform FAQs

98. Is CareBridge free to use?

The knowledge base does not state a specific pricing model for donors or orphanages; donation transactions themselves are not processed as payments by the platform (see Missing Information in Chapter 7).

99. What roles exist on CareBridge?

Three roles: DONOR, ORPHANAGE, and ADMIN, each with distinct permissions enforced by role-based access control in SecurityConfig.

100. Can an orphanage donate to another orphanage?

No. The orphanage feature matrix explicitly excludes browsing other orphanages or donating to them; orphanages post needs and receive donations rather than making them.

101. Can a donor post a need?

No. Posting needs is an orphanage-only capability; the donor feature matrix explicitly excludes it.

102. What happens to my uploaded documents if my profile is rejected?

Your submitted profile data is copied into the RejectedDonorProfile or RejectedOrpProfile table along with the rejection reason, and the original active profile record is deleted.

103. Does CareBridge have a mobile app?

Not currently. A mobile app (React Native or Flutter) is listed under the medium-term future roadmap, not as an existing feature.

104. Is there a WhatsApp or SMS notification option?

Not currently. WhatsApp integration and push notifications are listed under the near-term future roadmap; today, notifications are sent by email only.

105. Can I see all slots across every orphanage in one place?

The admin portal includes a GlobalSlotsPage.jsx for viewing all slots system-wide; regular donors browse slots per-orphanage via the available-slots endpoint.

106. What database does CareBridge use?

MySQL in production and H2 (in-memory) for testing, accessed through Spring Data JPA with Hibernate as the ORM.

107. Who can access the Swagger API documentation?

The /swagger-ui/** and /v3/api-docs/** routes are permitted without authentication, per the SecurityConfig public route list.

108. What happens if I try to book a slot with more visitors than the remaining capacity?

The booking is validated against remaining capacity server-side; a request for more visitors than are available will not be accepted.

109. Is my donation reserved quantity the same as my delivered quantity?

No. Reserved quantity reflects donations pledged but not yet delivered; fulfilled quantity reflects donations actually delivered and confirmed by the orphanage.

110. Can I see donor names before I approve a visit booking as an orphanage?

Yes. The pending-bookings view shown to orphanages includes donor details and visitor counts to support the confirm/reject decision.

111. What is the JJ Act Certificate?

It is documentary proof of an orphanage's compliance with India's Juvenile Justice Act, required alongside DARPAN ID and PAN during orphanage registration.

112. Does CareBridge support recurring donations?

Not currently. Recurring donation support is listed under the medium-term future roadmap rather than an existing feature.

113. Can I get a donation receipt or certificate?

Not currently. Donation receipts/certificates are listed under the near-term future roadmap rather than an existing feature.

114. What happens to Cloudinary-hosted images if I delete my account?

Images are permanent in Cloudinary and are not deleted automatically when a related profile is removed, since the database only stores the image URL, not the file itself.

115. Is two-factor authentication available?

Not yet. Two-factor authentication (2FA) is listed among security practices still to be implemented.

116. Can I change my registered email address?

The knowledge base does not document an email-change endpoint; this is flagged as a documentation gap in Chapter 7.

117. How is my subscription status used?

DonorProfile includes a subscriptionStatus field (active/inactive) and it is surfaced on the donor dashboard profile summary; the knowledge base does not detail what subscription tiers or billing exist, which is flagged as a gap in Chapter 7.

118. What Java version does the backend require?

Java 21, running Spring Boot 4.0.6 with Maven as the build tool.

119. What frontend build tool does CareBridge use?

Vite — version 8.0.10 for the main frontend and 8.0.12 for the admin portal, both paired with React 19.x and Bootstrap/React Icons for styling and iconography.

120. Is there a staging environment separate from production?

The knowledge base documents development (dev) and production (prod) configuration profiles (application-dev.properties, application-prod.properties) but does not mention a separate staging environment.

121. How large can an uploaded profile picture be?

Up to 20MB per file, with a 100MB cap on the total multipart request size.

122. Does CareBridge support dark mode?

The knowledge base does not document a dark mode option in the current frontend.

123. Can a rejected donor immediately re-register with the same email?

The knowledge base indicates rejection is permanent for that submission and requires re-registering with corrected information; whether the same email can be reused immediately is not explicitly documented.

124. What happens if two donors try to book the last remaining slot capacity simultaneously?

The knowledge base does not document specific concurrency-handling behavior for simultaneous slot bookings; this is flagged as a technical gap in Chapter 7.

125. Does the platform show a map view of orphanage locations?

The knowledge base documents location fields (village, mandal, district, state, country) but does not mention a map-based visualization feature.

126. Is there an API rate limit today?

No. API rate limiting is explicitly listed as a security practice still to be implemented, not one currently in place.

127. Can an orphanage see aggregate donor demographics?

The knowledge base does not document any demographic-aggregation feature for orphanages beyond individual donor details on specific donations and bookings.

Total FAQ entries in this chapter: 127, covering registration, authentication, profiles, verification, needs, donations, slots and visit booking, dashboards, notifications, messaging, landing/browse, admin operations, technical architecture, troubleshooting, and platform policy.

Chapter 7 — Knowledge Gap Analysis

This chapter identifies questions a real user of any persona could reasonably ask that the current CareBridge knowledge base cannot fully answer. Per the documentation brief, no gap below is filled with an invented fact — each is instead named, explained, and given a priority and a suggested next step. This chapter doubles as a product backlog input and as an explicit signal to a RAG chatbot about where it should say 'I don't have that information' rather than guessing.

7.1 Pricing & subscription model

Affects: Investors, Donors

Why it matters: DonorProfile includes a subscriptionStatus field, but the knowledge base never documents what a subscription unlocks, what tiers exist, or whether it is free.

What should be added: A pricing/plans page and a subscriptionTier enum with documented feature gating.

Priority: High

Suggested implementation: Add a Plans entity and expose it via a dedicated /api/plans endpoint; document the mapping from tier to feature access.

7.2 Payment processing

Affects: Donors, Investors, NGOs

Why it matters: The platform explicitly does not process payments today, which is a major gap for donors who would prefer a single-click online contribution.

What should be added: Integration with a payment gateway (e.g., Razorpay, already flagged on the roadmap) and supporting transaction records.

Priority: High

Suggested implementation: Introduce a PaymentTransaction entity linked to DonationRequest and a webhook handler for gateway confirmation.

7.3 Donor-to-donor visibility on shared needs

Affects: Donors

Why it matters: Multiple donors can fund the same need item, but the knowledge base does not state whether donors can see each other's contributions or names.

What should be added: A documented policy (visible progress bar with anonymized aggregate contributor count, or full visibility).

Priority: Medium

Suggested implementation: Add a contributorsCount or contributorsSummary field to the NeedItemResponse DTO.

7.4 Email address change

Affects: Donors, Orphanages

Why it matters: No endpoint for changing a registered email is documented, which is a common account-management request.

What should be added: A documented email-change flow requiring re-verification of the new address.

Priority: Medium

Suggested implementation: Add POST /auth/change-email with OTP confirmation to both old and new addresses.

7.5 Real-time messaging behavior

Affects: Donors, Orphanages, Developers

Why it matters: Messaging components exist (Messages.jsx and role-specific pages) but the knowledge base does not confirm whether delivery is real-time, polling-based, or requires a manual refresh.

What should be added: Explicit documentation of the messaging transport, and a roadmap note already references WebSockets as a near-term addition, implying the current state may be non-real-time.

Priority: Medium

Suggested implementation: Document current polling interval if applicable; prioritize WebSocket upgrade per the existing roadmap.

7.6 Dashboard refresh cadence

Affects: Donors, Orphanages

Why it matters: Neither dashboard's refresh interval or caching policy is documented, leaving it unclear whether users see live or stale data after an action.

What should be added: A documented cache TTL or explicit note that dashboards query live on each load.

Priority: Low

Suggested implementation: Add a lastUpdated timestamp field to dashboard responses so the frontend can display data freshness.

7.7 Multiple bookings on the same day

Affects: Donors, QA Engineers

Why it matters: The knowledge base does not state whether a donor can hold two separate confirmed bookings for the same date, which matters for both UX and testing.

What should be added: An explicit business rule, enforced server-side if a restriction is intended.

Priority: Low

Suggested implementation: Add a validation check in DonorSlotService if same-day duplicate bookings should be blocked.

7.8 Accessibility compliance

Affects: Accessibility Experts

Why it matters: No accessibility standard (e.g., WCAG 2.1 AA) is referenced anywhere in the knowledge base for either the donor/orphanage frontend or the admin portal.

What should be added: A documented accessibility target and an audit against it, particularly given the platform serves a wide range of donor ages and technical literacy.

Priority: High

Suggested implementation: Run an automated audit (axe-core) plus manual screen-reader testing on the core donation and registration flows.

7.9 Data retention & deletion policy (GDPR-style)

Affects: Security Experts, Government Organizations

Why it matters: Security best practices explicitly list 'GDPR compliance (data deletion)' as not yet implemented, and no retention schedule for archived/rejected profiles is documented.

What should be added: A documented data-retention and right-to-deletion policy, especially for RejectedDonorProfile and RejectedOrpProfile archives containing PAN data.

Priority: High

Suggested implementation: Define a retention period for archive tables and build a deletion-request workflow.

7.10 Rate limiting & abuse prevention

Affects: Security Experts, Developers

Why it matters: API rate limiting is explicitly listed as 'to implement' rather than in place, leaving OTP and login endpoints potentially exposed to brute-force or spam abuse.

What should be added: Rate limiting on /auth/** and OTP-related endpoints in particular.

Priority: High

Suggested implementation: Add a token-bucket rate limiter (e.g., Bucket4j) in front of authentication and OTP endpoints.

7.11 Audit logging of admin actions

Affects: Security Experts, Platform Admin

Why it matters: Approvals and rejections are described as 'logged in database' at a high level, but no structured audit-log schema (who, what, when, before/after state) is documented.

What should be added: A dedicated AuditLog entity capturing every admin decision with before/after state.

Priority: Medium

Suggested implementation: Introduce an `@EntityListener` or AOP-based audit logger around `AdminService` methods.

7.12 Donor and orphanage identity documents beyond PAN

Affects: NGOs, Government Organizations

Why it matters: Only PAN is required for donor identity; no secondary ID (Aadhaar, passport) is documented as an option, which may limit verification confidence for high-value donors.

What should be added: An optional secondary-ID upload path for higher-trust donor tiers.

Priority: Low

Suggested implementation: Add an optional `secondaryIdType/secondaryIdPhoto` pair to `DonorProfile`.

7.13 Volunteer & non-monetary contribution tracking

Affects: NGOs, Orphanages

Why it matters: The platform currently tracks only item/monetary donations and visits; skill-based volunteering (teaching, medical camps) is not modeled anywhere.

What should be added: A volunteer module, already noted as a long-term roadmap item.

Priority: Medium (roadmap)

Suggested implementation: Model a `VolunteerOpportunity` entity parallel to `NeedItem`, with its own booking flow.

7.14 Automated testing coverage detail

Affects: QA Engineers, Developers

Why it matters: Testing tools (Mockito, H2, JUnit, ESLint, Playwright/Cypress) are named, but no coverage percentage, CI pipeline, or test-suite breakdown is documented.

What should be added: A documented CI pipeline (build, test, lint stages) with coverage thresholds.

Priority: Medium

Suggested implementation: Publish a coverage badge and CI configuration (e.g., GitHub Actions) alongside the repository.

7.15 Internationalization & multi-language support

Affects: NGOs, Government Organizations

Why it matters: Multi-language support is listed only as a long-term roadmap item; the current UI language is not documented but implied to be English-only.

What should be added: `i18n` scaffolding (e.g., `react-i18next`) ahead of full translation.

Priority: Low (roadmap)

Suggested implementation: Externalize UI strings into locale files as groundwork before translation work begins.

Chapter 8 — RAG & Qdrant Optimization Guide

CareBot is built on Retrieval-Augmented Generation (RAG): an OpenRouter-hosted LLM generates answers, grounded by passages retrieved from a Qdrant Cloud vector database populated with embeddings of this documentation. This chapter gives concrete, implementation-level guidance for getting the most reliable retrieval quality out of that pipeline, using Chapters 5 and 6 of this very document as the primary source corpus.

8.1 Chunking Strategy

Each Chapter 5 topic entry (question, answer, metadata) is already an ideal retrieval unit — self-contained, single-intent, and short enough to fit comfortably inside typical embedding context windows. Rather than chunking by fixed token count, chunk by topic boundary: one chunk equals one topic's canonical question plus its answer plus its metadata block. This avoids the common RAG failure mode of a chunk boundary splitting a question from its answer.

- Target chunk size: roughly 150–300 tokens per topic entry (canonical Q + answer).
- Avoid chunk overlap for Chapter 5/6 entries specifically, since each entry is already atomic; overlap is more useful for the narrative chapters (1–4, 7–10) where context spans paragraphs.
- For narrative chapters, use a 200–300 token chunk size with roughly 15–20% overlap so a fact split across a paragraph boundary is still retrievable from either chunk.

8.2 Metadata Design

Every point stored in Qdrant should carry a payload beyond the raw text, enabling filtered and hybrid retrieval rather than pure semantic search alone.

- **category** (e.g., 'Donations', 'Visit Slots & Booking') for topic-scoped filtering.
- **persona** (New Visitor, Donor, Orphanage Admin, Platform Admin, Developer) to bias retrieval toward the asking user's likely role.
- **difficulty** (Basic, Intermediate, Advanced) to avoid surfacing developer-level detail to a first-time donor.
- **topic_id** (e.g., 'DON-01') for exact-match lookups and for resolving 'related_topics' cross-references.
- **source_chapter** and **last_verified_date** so answers can be traced back to this document and flagged if the source changes.

8.3 Collection Design in Qdrant

A single collection (e.g., `carebridge_kb`) with payload-based filtering is sufficient at this corpus size; splitting into per-category collections adds operational overhead without a retrieval-quality benefit until the corpus grows substantially larger. Configure the vector field with cosine distance (standard for normalized sentence-embedding models) and enable payload indexing on `category`, `persona`, and `topic_id` for fast

filtered queries.

8.4 Embedding Strategy

- Embed the canonical question and answer together as one string per topic, since user queries more closely resemble questions than answers, and concatenation keeps both signals in the same vector.
- Additionally embed each alternative phrasing (Chapter 5's variant questions) as separate points sharing the same topic_id in payload, so a typo'd or voice-search-style query still lands near its canonical answer.
- Re-embed the corpus whenever this documentation is revised; stale embeddings against updated source text are a common, hard-to-detect source of hallucination.

8.5 Hybrid Search & Re-ranking

Pure vector similarity search can miss queries that hinge on an exact term (a specific endpoint path, an exact status name like 'DELIVERED', or 'CareBridgeID'). Combine dense vector search with a sparse/BM25-style keyword pass over the 'keywords' metadata field, then merge and re-rank the combined result set (for example, weighting exact keyword matches higher) before passing the top few passages to the LLM.

8.6 Context Retrieval & Conversation Memory

- Retrieve the top 3–5 passages per query rather than 1, since CareBot conversations often require combining a primary answer with a related clarification (use the 'related_topics' field to pull those in deliberately).
- Carry the last 2–3 conversation turns as retrieval context, not just the latest message, so follow-up questions like 'and what if it's rejected?' resolve correctly against the prior topic.
- Reset conversation memory between unrelated topic switches to avoid stale context leaking into an unrelated answer.

8.7 Hallucination Prevention

- Instruct the LLM to answer only from retrieved passages and to explicitly say 'I don't have that information yet' when no passage covers the question — Chapter 7's gap list is the canonical source of what to say this about.
- Never let the model fill a gap with a plausible-sounding but unverified answer about PAN formats, statuses, or endpoints; these are exactly the kinds of specific facts users will later act on.
- Log queries that return low-confidence retrieval scores for human review; this surfaces real gaps that Chapter 7 may have missed.

8.8 Prompt Engineering for CareBot

System prompts for CareBot should state its scope explicitly (CareBridge platform questions only), instruct it to cite which feature/module an answer relates to, ask it to keep answers concise for a chat UI (2–4

sentences, expandable on request), and instruct it to never expose internal identifiers like a raw database id, JWT secret, or SMTP credential even if a retrieved passage were to contain one.

8.9 Retrieval Optimization Checklist

- Evaluate retrieval quality with a held-out test set built from Chapter 5's alternative phrasings (they were generated for exactly this purpose).
- Track precision@3 and recall@5 against that test set after every embedding-model or chunking change.
- Periodically prune or merge near-duplicate topic entries as new documentation is added, to avoid diluting retrieval scores across redundant chunks.

Chapter 9 — Future Improvements

CareBridge's own knowledge base already documents a phased roadmap. This chapter restates that roadmap with the rationale and expected benefit of each item, and cross-references it against the gaps identified in Chapter 7 where relevant.

9.x Near Term

Push notifications (FCM for mobile). Immediate, app-level alerts for approvals, donation status changes, and booking decisions, reducing reliance on email alone.

WhatsApp integration for messages. Meets donors and orphanages on a channel many already use daily in the Indian market.

Real-time chat (WebSockets). Upgrades the existing Messages.jsx experience from likely-polling to true real-time delivery.

Advanced search and filters. Extends today's category/priority filtering with location radius, fulfillment percentage, and free-text search.

Donation receipts/certificates. Gives donors a shareable, presentable artifact of their contribution beyond dashboard statistics.

9.x Medium Term

Mobile app (React Native/Flutter). Extends the current web-only frontend to native mobile, important given many donors are mobile-first.

Payment integration (Razorpay). Directly addresses the Chapter 7 gap around the platform not currently processing payments.

Impact reports and analytics. Deeper, exportable reporting beyond the current dashboard summaries.

Recurring donation support. Lets donors commit to a need on an ongoing schedule rather than a single transaction.

Gift cards. Enables third parties to fund a donor's giving without needing their own verified profile.

9.x Long Term

AI-powered need recommendations. A more sophisticated matching layer than today's 'Urgent Needs' dashboard section.

Blockchain for transparency. An immutable donation ledger, extending the platform's existing transparency-first design philosophy.

Corporate donations platform. A dedicated flow for organizations donating at scale, likely with different verification requirements than individual donors.

Volunteer management. Addresses the Chapter 7 gap around non-monetary, skill-based contributions.

Fundraising campaigns. Time-boxed, goal-based campaigns spanning multiple need items.

Multi-language support. Addresses the Chapter 7 internationalization gap.

Video verification for profiles. A stronger identity signal than static document photos alone.

Chapter 10 — Technical Overview

This chapter consolidates the technical architecture of CareBridge at a level appropriate for engineers joining the project, without exposing any credentials, secrets, or environment-specific configuration values.

10.1 Architecture Overview

CareBridge is a modular-monolith backend (Spring Boot 4.0.6, Java 21, Maven) exposing a REST API consumed by two independent React single-page applications — the donor/orphanage frontend (React 19.2.5, Vite 8.0.10, Bootstrap 5.3.8) and the admin portal (React 19.2.6, Vite 8.0.12) — plus a public, unauthenticated landing-data surface for pre-registration visitors.

10.2 Module Breakdown

- authManagement — registration, login, OTP, JWT issuance
- donorManagement — donor profile CRUD and verification state
- orphanageManagement — orphanage profile CRUD and verification state
- donationManagement — donation lifecycle for both donor and orphanage sides
- visitbookingManagement — slot creation, booking, confirmation, expiry
- needsManagement — need item CRUD and fulfillment tracking
- dashBoardManagement — donor and orphanage analytics aggregation
- notificationManagement — OTP and email notification delivery
- adminManagement — profile approval/rejection workflows
- slotDetailsFetchManagment — combined orphanage+slot detail queries
- landingPagedata — public statistics and browse endpoints
- common — Cloudinary config, image upload service, shared API response wrapper

10.3 Backend Overview

Each module follows the same internal layering: a Controller exposes REST endpoints, a Service holds business logic, a Repository (Spring Data JPA) handles persistence, and Entities map to MySQL tables via Hibernate. DTOs mediate every request and response so entity internals are never exposed directly over the wire, and module-specific custom exceptions provide precise error signaling (e.g., DonorProfileNotFoundException, OtpExpiredException).

10.4 Frontend Overview

The donor/orphanage frontend organizes code into 13 API integration files, 33+ reusable components, and 41+ pages, with a dedicated chatbot subdirectory (CareBot.jsx, ChatWindow.jsx, ChatMessage.jsx, ChatInput.jsx, SuggestedQuestions.jsx, TypingIndicator.jsx) and route-level guards separating public, donor, orphanage, and admin-protected routes.

10.5 Admin Portal

A structurally separate React application with its own API layer (apiConfig.js, authApi.js, donorApi.js, orphanageApi.js), dedicated pages per domain area (donations, donors, orphanages, settings, visits), and shared components like AdminSidebar.jsx and RejectModal.jsx for the approve/reject workflow.

10.6 Authentication Flow

Registration and login both funnel through the authManagement module: OTP-gated registration creates a User record with a hashed password and a role; login issues a 24-hour JWT; every subsequent request passes that JWT in an Authorization header, which JwtAuthFilter validates before SecurityConfig's role rules decide whether the request may proceed.

10.7 Database Design

Twelve core entities and three archive tables model the domain: User, DonorProfile, OrphanageProfile, NeedItem, DonationRequest, Slot, VisitBooking, Otp, RejectedDonorProfile, RejectedOrpProfile, and DeletedItems (see Chapter 2 for the full entity-relationship description). Foreign keys use each profile's CareBridgeID as the natural cross-module reference rather than raw database primary keys.

10.8 API Organization

Endpoints are grouped by role prefix: /auth/** is unauthenticated, /donor/** requires ROLE_DONOR, /orphanage/** requires ROLE_ORPHANAGE, /admin/** requires ROLE_ADMIN, and /api/landing/** is public. Swagger/OpenAPI documentation is generated automatically via springdoc-openapi-starter-webmvc-ui and served at /swagger-ui/** and /v3/api-docs/**.

10.9 Deployment Overview

The backend builds via Maven into a runnable jar deployable to any Java 21-compatible cloud host with a MySQL instance and the required environment variables (database credentials, JWT secret, SMTP credentials, Cloudinary keys, CORS origins — none of which are reproduced in this document). Both frontend applications build via 'npm run build' and deploy independently to static hosts such as Firebase Hosting, Vercel, or Netlify.

10.10 Scalability & Maintainability

- Stateless JWT authentication allows horizontal scaling of backend instances without shared session state.
- Indexed lookups on email, phone, and CareBridgeID keep frequent queries fast as data volume grows.
- The documented near-term improvements — Redis caching, pagination on list endpoints, and API rate limiting — directly target scalability before they become production incidents.

10.11 Security Overview

See Chapter 3.10 for the full feature-level treatment and Chapter 7 for security gaps still to close. In summary: BCrypt password hashing, stateless JWT auth, role-based access control, CORS restrictions, input validation, and OTP expiry/single-use are implemented today; HTTPS enforcement, rate limiting, 2FA, audit logging, encryption at rest, and GDPR-style deletion are not yet.

10.12 Testing Strategy

Backend testing uses Mockito for unit tests and an H2 in-memory database for integration and repository tests, run via 'mvn test' (or 'mvn verify' for the full suite). Frontend testing uses React Testing Library for components and Playwright or Cypress for end-to-end coverage, supplemented by manual cross-browser checks (Chrome, Firefox, Safari). ESLint enforces frontend code quality; SonarQube is listed as an optional addition for deeper static analysis.

Appendix A — Complete API Endpoint Reference

The following tables enumerate every REST endpoint documented in the CareBridge knowledge base, grouped by access level. This is intended as a fast-lookup technical reference; see Chapter 3 for narrative feature documentation of the same functionality.

Public Endpoints (No Authentication Required)

Method	Endpoint	Description
GET	/api/landing/stats	Global platform statistics
GET	/api/landing/orphanages	Browse all verified orphanages
GET	/api/landing/needs	Browse all active needs
GET	/api/landing/features	Platform feature descriptions
GET	/swagger-ui.html	API documentation UI
GET	/v3/api-docs	OpenAPI machine-readable spec

Authentication Endpoints

Method	Endpoint	Description
POST	/auth/register	Start registration (sends email OTP)
POST	/auth/verify	Verify OTP and complete registration
POST	/auth/resend	Resend registration OTP
POST	/auth/login	Login and receive a 24-hour JWT
POST	/auth/forgot-password	Start password reset
POST	/auth/verify-forgot-otp	Verify password-reset OTP
POST	/auth/reset-password	Set a new password

Donor Endpoints (ROLE_DONOR)

Method	Endpoint	Description
POST	/donor/create-profile	Create donor profile (multipart)
PUT	/donor/update-profile	Update donor profile (multipart)
GET	/donor/{id}	Get donor details
POST	/donor/donate	Create a donation (multipart)

Method	Endpoint	Description
GET	/donor/pending-donations	View pending donations
GET	/donor/completed-donations	View completed donations
GET	/donor/cancelled-donations	View cancelled donations
GET	/donor/delivered-donations	View delivered donations
PATCH	/donor/update/{donationId}	Update a pending donation (multipart)
DELETE	/donor/delete-pending-donation/{id}	Delete a pending donation
GET	/donor/slots/available-slots/{orphanageId}	Browse available slots
POST	/donor/slots/book	Book a slot
GET	/donor/allNeeds	Browse all active needs
GET	/api/browse	Search and filter needs/orphanages
GET	/donor/dashboard	Donor dashboard statistics

Orphanage Endpoints (ROLE_ORPHANAGE)

Method	Endpoint	Description
POST	/orphanage/create-profile	Create orphanage profile (multipart)
PUT	/orphanage/update-profile	Update orphanage profile (multipart)
POST	/orphanage/verifyOrpOTP	Verify OTP during signup
POST	/orphanage/resendOrpOTP	Resend orphanage OTP
GET	/orphanage/{id}	Get orphanage details
POST	/orphanage/needItem/create-item	Create a need item
PUT	/orphanage/needItem/updateItem/{id}	Update a need item
DELETE	/orphanage/needItem/deleteItem/{id}	Delete a need item
GET	/orphanage/needItem/activeNeeds	View own active needs
GET	/orphanage/pending-donations	View pending donations received
GET	/orphanage/delivered-donations	View delivered donations
GET	/orphanage/cancelled-donations	View cancelled donations
POST	/orphanage/delivered/{id}	Confirm donation delivery (multipart)
POST	/orphanage/slot/create	Create a visit slot
GET	/orphanage/slot/pending	View pending visit bookings
PATCH	/orphanage/slot/confirm/{id}	Confirm a visit booking

Method	Endpoint	Description
PATCH	/orphanage/slot/reject/{id}	Reject a visit booking
GET	/orphanage/dashboard	Orphanage dashboard statistics

Admin Endpoints (ROLE_ADMIN)

Method	Endpoint	Description
GET	/admin/donor/pending	List pending donor profiles
GET	/admin/donor/approved	List approved donors
GET	/admin/donor/rejected	List rejected donors
PATCH	/admin/donor/approve/{id}	Approve a donor profile
PATCH	/admin/donor/reject/{id}	Reject a donor profile
GET	/admin/orphanage/pending	List pending orphanage profiles
GET	/admin/orphanage/approved	List approved orphanages
GET	/admin/orphanage/rejected	List rejected orphanages
PATCH	/admin/orphanage/approve/{id}	Approve an orphanage profile
PATCH	/admin/orphanage/reject/{id}	Reject an orphanage profile
GET	/admin/dashboard	Admin dashboard statistics

Utility Endpoints

Method	Endpoint	Description
POST	/notify/send-otp	Send OTP manually
GET	/slot/details/{slotId}	Get slot details
GET	/orphanage/{id}/slots	Get orphanage and its slots combined

Appendix B — Glossary of Terms

Definitions for CareBridge-specific terms and general technical terms as used throughout this documentation.

Term	Definition
CareBridgeID	A unique, system-generated identifier assigned to every donor and orphanage profile, used as the reference key
DARPAN ID	A registration number issued under India's social-sector registration scheme; required from every orphanage as p
JJ Act Certificate	Documentary proof of an orphanage's compliance with India's Juvenile Justice Act, required alongside DARPAN
PAN	Permanent Account Number — a 10-character Indian tax identifier ([A-Z]{5}[0-9]{4}[A-Z]{1}) required from both d
OTP	One-Time Password — a six-digit, single-use, time-boxed (5-minute) code emailed for registration, password res
JWT	JSON Web Token — the stateless bearer token issued at login (24-hour validity) used to authorize every protecte
RBAC	Role-Based Access Control — the mechanism restricting endpoints to ROLE_DONOR, ROLE_ORPHANAGE, or
Need Item	A specific, quantified requirement posted by an orphanage (category, quantity, price per unit, priority) that donors
Slot	A time window an orphanage makes available for donor visits, with a defined date, time range, and visitor capaci
Visit Booking	A donor's reservation against a specific slot, subject to orphanage confirmation or rejection.
DTO	Data Transfer Object — a class used to shape request/response payloads distinctly from internal database entitie
Cloudinary	The third-party cloud image-storage service used for all file uploads (profile pictures, documents, donation/deliver
CareBot	CareBridge's AI chatbot, built on a Retrieval-Augmented Generation (RAG) pipeline using an OpenRouter-hosted
RAG	Retrieval-Augmented Generation — an AI architecture in which a language model's answer is grounded by pass
Qdrant	The vector database used to store and search embeddings of CareBridge's documentation for CareBot's retrieva

Appendix C — Enum & Status Reference

A consolidated reference of every enumerated type used across CareBridge's data model, useful both for developers and for the CareBot retrieval corpus when a user asks 'what are all the possible statuses for X'.

Enum	Values
CategoryType (Need Items)	FOOD, EDUCATION, HEALTH, CLOTHING, TOYS, SPORTS, TECHNOLOGY, OTHER
PriorityLevel (Need Items)	LOW, MEDIUM, HIGH, URGENT
DonationType	IN_PERSON, ONLINE_ORDER
DonationStatus	PENDING, DELIVERED, CANCELLED (COMPLETED noted as a possible alternate status)
SlotStatus	AVAILABLE, EXPIRED, CANCELLED, FULL
VisitBookingStatus	PENDING, CONFIRMED, REJECTED, CANCELLED
Profile Verification Status (Donor & Orphanage)	PENDING, VERIFIED, REJECTED
User Role	DONOR, ORPHANAGE, ADMIN
ActivityType (Orphanage Dashboard)	DONATION_RECEIVED, VISIT_SCHEDULED, NEED_POSTED, and related activity-feed types

Appendix D — Core Entity Field Reference

Key fields for the primary database entities, consolidated from Chapter 2's entity-relationship description for quick lookup.

User

Field
id (PK)
name
email (unique)
passwordHash
role
isEmailVerified
isActive
createdAt
updatedAt

DonorProfile

Field
id (PK)
donorId (FK→User)
careBridgeID (unique)
name
dob
designation
phone (unique)
address fields
panNumber (unique)
panPhoto
profilePic
donorStatus
subscriptionStatus

OrphanageProfile

Field
id (PK)
orphanageId (FK→User)
careBridgeID (unique)
orphanageName
adminName
numberOfChildren
email (unique)
phone (unique)
darpanId
panNumber
panPhoto
jjActCertificatePhoto
address fields
orphanageStatus
emailVerified

NeedItem

Field
id (PK)
needItemId (unique)
orphanageCareBridgeId (FK)
name
description
category
quantity
pricePerQuantity
priority
fulfilledQuantity
reservedQuantity
isActive

Field
productLinks
createdAt
deadline

DonationRequest

Field
id (PK)
donationRequestId (unique)
donorCareBridgeId (FK)
orphanageCareBridgeId (FK)
needItemId (FK)
quantity
totalValue
donationType
donationStatus
expectedVisitDateTime
expectedDeliveryDate
actualDeliveryDate
platformName
trackingId
orderProofImageUrl
deliveryNote
deliveryPhotoUrl
message
createdAt

Slot

Field
id (PK)
slotId (unique)
orphanageCareBridgeId (FK)

Field
slotDate
startTime
endTime
capacity
reservations
slotStatus
createdAt

VisitBooking

Field
id (PK)
bookingId (unique)
slotId (FK)
donorCareBridgeId (FK)
numberOfVisitors
bookingStatus
visitorMessage
rejectionReason
createdAt
confirmedAt
rejectedAt

Otp

Field
id (PK)
email
otp (6-digit)
expiryTime
usedFlag
createdAt

Appendix E — Feature Matrix by Role

A consolidated view of what each role can and cannot do on CareBridge.

Capability	Donor	Orphanage	Admin
Register & create profile	Yes	Yes	N/A
Browse needs & orphanages	Yes	No	N/A
Post needs	No	Yes	No
Donate items	Yes	No	No
Create visit slots	No	Yes	No
Book a visit	Yes	No	No
Confirm/reject visit bookings	No	Yes	No
Confirm donation delivery	No	Yes	No
Approve/reject profiles	No	No	Yes
View platform-wide statistics	No	No	Yes
Track own impact/fulfillment	Yes	Yes	N/A
Messaging	Yes	Yes	N/A

Appendix F — Version History

This documentation reflects the state of the CareBridge knowledge base as of its preparation date. As CareBridge evolves, this document should be regenerated from an updated knowledge base rather than hand-edited, to keep it consistent with the codebase and safe for continued use as a RAG training corpus.

Version	Date	Notes
1.0	July 2026	Initial comprehensive documentation and AI knowledge base release.

Appendix G — Common HTTP Status Codes on CareBridge

A quick reference for the HTTP status codes a client integrating with the CareBridge API is most likely to encounter, and what each one means in context.

Code	Meaning	Typical CareBridge Cause
200 OK	Request succeeded	Standard successful GET/PATCH response
201 Created	Resource created	Successful profile, donation, need, or slot creation
400 Bad Request	Validation failure	Missing required field, malformed PAN, invalid enum value
401 Unauthorized	Authentication failure	Missing, invalid, or expired JWT
403 Forbidden	Authorization failure	Valid JWT but wrong role for the requested endpoint
404 Not Found	Resource not found	Unknown donor/orphanage/need/slot/booking id
409 Conflict	Uniqueness violation	Duplicate email, phone, or PAN number
413 Payload Too Large	Upload limit exceeded	File over 20MB or request over 100MB
500 Internal Server Error	Unhandled server error	Unexpected exception; should be rare and logged

For persona-specific guidance on interpreting these codes in the CareBridge context, see the Troubleshooting section of Chapter 6 and the Common Issues section of Chapter 10.

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Conclusion

CareBridge is a comprehensive platform designed to streamline charitable giving through verified profiles, transparent need posting, and structured interactions between donors and orphanages.

Key Strengths

- Role-based separation (Donor, Orphanage, Admin)
- Comprehensive verification process
- Multiple donation modes (in-person and online)
- Structured visit booking system
- Impact tracking and transparency
- Scalable architecture

Key Values

- Accountability: Admin verification, document requirements
- Transparency: Need posting, donation tracking, impact metrics
- Efficiency: Matching donors with actual needs
- Trust: Profile verification, rejection tracking
- User Experience: Multiple interfaces (Donor, Orphanage, Admin)

The system is production-ready but can benefit from additional features like real-time notifications, advanced analytics, and mobile support, as detailed in Chapter 9.

End of CareBridge Documentation & AI Knowledge Base.